

Understanding Basket-Shopping Consumer Behavior Using Causal Product Networks

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Problem definition: Most retail merchandise planning is organized by product categories, yet retail demand is realized through shopping baskets. We study the problem of utilizing observational basket-shopping data to measure causal within-basket spillovers, which are critical for retail decision-making and for understanding how the purchase of one product affects purchases of others. **Methodology:** We propose a novel causal product network (CPN)-based estimation model, where products are represented as nodes in a directed network, with edges capturing within-basket spillover effects, to jointly capture cross-category and within-category purchases in customer baskets. We apply this model to empirically study basket-shopping behavior using causal structure learning on transaction-level grocery basket data. **Results:** We show that the proposed CPN-based demand model represents the interactions among product purchases in a shopping basket more accurately than alternative theory- and machine-learning-driven network structures and yield a model with more *parsimonious* and tractable set of product-specific relationships. Building on the CPN framework, we also find that product-level specifications represent consumer behavior significantly more accurately than the commonly assumed category-level specifications. Finally, we investigate how within-basket spillovers vary across shopping channels. We find that relative to online shopping, brick-and-mortar baskets exhibit systematically denser spillover links among product purchases. **Managerial implications:** Our paper provides retailers with a method and empirical rationale to filter out spurious correlations and identify within-basket product spillovers for their operational decision-making. Furthermore, retailers should account for cross-channel differences and estimate these relationships at the product level rather than the category level.

Key words: basket shopping, product network, causal structure learning, empirical research

1. Introduction

Measuring how the purchase of one product influences the purchase of another within the same shopping basket is important for retailers' operational decision-making. However, because modern assortments include a large number of products, identifying these within-basket effects at scale is a challenging problem. In this paper, we are motivated by recent advances in causal machine learning, in particular causal structure learning, which enable the estimation of directional cross-item effects using observational basket data. Such an approach can be extremely beneficial because it can be

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applied at scale to measure cross-item relationships among product purchases, and it can be utilized to extract value from the vast amounts of basket-shopping data that are available to retailers. Thus, building on this literature, we introduce Causal Product Networks (CPNs), a directed-graph representation of within-basket spillovers across and within categories that focuses on direct cross-item effects. Using the CPN framework, we address the following research questions: (1) How do causal product networks perform in capturing the relationships among product purchases compared to alternative methods for characterizing basket-shopping behavior, including both theory-driven and machine-learning-based approaches? (2) When balancing model complexity and model fit, does a model with product-level causal effects, in which the causal effects between individual products are allowed to differ, outperform more parsimonious specifications, such as a category-dependent model in which all causal effects from products in one category to products in another category are constrained to be identical? (3) Do product purchase relationships in basket shopping vary across different retailing formats? Specifically, do these relationships differ across the brick-and-mortar (B&M) and online channels?

To study these questions, we collaborate with Numerator, a prominent market research company known for its first-party, consumer-sourced data. Numerator gathers purchase data for a large number of consumers across various B&M and online channels; transaction sources include retailer loyalty data, mobile app purchases, email as well as paper receipts. This dataset allows us to test our hypotheses across different shopping contexts. Specifically, we focus on five grocery themes—pasta, quick-service restaurant, bakery, prepared food, and unprepared food within the Numerator dataset. A theme is defined as a collection of related product categories that together represent different shopping contexts, with each category containing multiple products. We estimate CPNs and alternative methods separately for each theme to examine whether our findings generalize across different themes, thereby providing evidence on basket-shopping behavior across a diverse range of product types and shopping patterns.

Our empirical analysis provides consistent support for the hypotheses corresponding to all the research questions. First, we find that CPNs, which capture causal relationships among product purchases, represent product purchase relationships in shopping baskets more accurately than several alternative approaches, including (1) Lasso- and Ridge-regularized complete networks, (2) singular value decomposition (SVD)- and eigenvalue decomposition (EVD)-based correlation networks, (3) collaborative filtering-based networks that connect items to their nearest neighbors in co-occurrence space, and (4) association networks based on Apriori-style association rules from the seminal work of Agrawal et al. (1993). To estimate CPNs, we use causal structure learning, which learns relationships from observational data without the need for expensive randomized experiments (Pearl 1988, Spirtes et al. 2000a). This method is well suited to basket data, where many products across

categories are jointly observed in the same transaction but experimentation is limited. Importantly, causal structure learning goes beyond co-purchase patterns by leveraging conditional independence to filter out associations that are mediated by other products in the basket, retaining only direct spillover effects. Across the five themes and both channels, CPN-based models achieve systematically better out-of-sample fit while using substantially fewer parameters, indicating that they capture the dependence structure more parsimoniously. Specifically, based on the Akaike Information Criterion (AIC), the CPN-based models consistently exhibit significantly lower AIC values than the regularized complete, correlation-based, collaborative filtering-based, and association-based benchmark models. These findings provide strong evidence that a CPN structure learned from conditional dependence patterns offers a more suitable representation of within-basket relationships in our data than the alternatives.

Second, building on the CPN-based demand model, we compare a range of alternative category-level and product-level specifications that have been assumed in the literature and are constructed by imposing different assumptions on cross-product demand effects. We find that, even after accounting for model complexity, allowing spillover effects to vary at the product-level represents basket-shopping behavior substantially better than category-level restrictions that force all interactions within a category pair to be the same. This result shows that consumers’ basket-shopping behavior is driven by product-specific spillovers, not category-level interactions as assumed in the prior literature. For example, we find that the categories of condiments and beverages are statistically independent, but purchasing fruit juice, a product in the beverages category, can affect the purchase of salad dressing, a product in the condiments category. Furthermore, spillovers within the same pair of categories exhibit substantial product-level heterogeneity in both magnitude and direction, implying that category-level aggregation can be misleading even when effects do not simply offset one another. Thus, product-level effects provide a more accurate description of basket-shopping consumer behavior. Moreover, we find that the number of edges in the product-level causal model is surprisingly only marginally larger than that in the alternative restriction-based specifications, because the causal structure learning procedure screens out many unsupported edges. This sparsity is important given the potentially very large number of parameters in basket-level demand models, and it suggests that considering product-level demand does not require a much complex model.

Third, we investigate how within-basket spillovers vary across different types of retail settings. Comparing online and B&M shopping baskets, we find that the online shopping channel exhibits fewer causal relationships among product purchases within shopping baskets compared to the B&M channel, and that the structure and sign of these relationships differ systematically across channels. In physical stores, causal spillovers exhibit greater density, and links more frequently involve perishable and impulse-oriented items. This pattern is consistent with in-store meal planning and

spontaneous purchasing behavior. In contrast, we find that online shoppers tend to add items to their baskets more independently of other purchases, with online cross-item links concentrated in staple and replenishment products. These differences illustrate how CPNs reveal channel-specific patterns in basket formation and show that spillovers are channel-dependent, with direct implications for basket-aware category management and cross-category coordination.

Our paper has important implications for retailers. First, we demonstrate the application of a causal structure learning approach to basket-shopping data to uncover causal relationships among product purchases. Second, we empirically show that the resulting causal structure is both parsimonious and better fitted to capture cross-product demand effects. Third, our study provides empirical evidence that cross-product demand effects vary across different channels, suggesting that channel-specific knowledge can inform retail decision-making. Taken together, our findings demonstrate the value of causal structure learning for characterizing basket-shopping behavior and for providing insights into relationships among product purchases. The contribution of our paper is to introduce a new representation of basket-shopping behavior as a causal product network, to evaluate this representation rigorously against leading alternatives, and to generate new insights for modeling customer choice in retail settings.

The remainder of this paper is organized as follows: §2 summarizes the related literature; §3 presents our causal product network model and develops the hypotheses regarding basket-shopping purchase decisions; §4 describes the empirical setting and the data; §5 discusses the estimation method; §6 presents the empirical results; and §7 concludes the paper.

2. Literature Review

Our paper is related to the existing literature on (1) utility-based and data-driven models of choice and basket-shopping behavior, (2) applications of basket-shopping data in operational decisions, and (3) causal structure learning. In this section, we review the relevant literature on these topics and describe the contributions of our work.

Choice modeling for basket-shopping customers seeks to discover and represent the complementary and substitution effects among product purchases with the goal of facilitating optimal decisions for product variety, pricing, and inventories. One common approach relies on the multinomial logit model (MNL), which is widely used in discrete choice modeling to capture substitution behavior within a category for either a single utility-maximizing choice (Börsch-Supan 1990, McFadden and Train 2000) or multiple purchase decisions (Aouad and Segev 2023, Bai et al. 2023, Aouad et al. 2024). The traditional MNL framework has been extended in the recent years to model multiple-purchase decisions across categories in basket shopping. For example, Cachon and Kök (2007) extend the nested-MNL model to describe the behavior of customers who choose one product from each of

two categories and to demonstrate its performance in an assortment optimization problem. Jasin et al. (2023) develop the Multivariate Multinomial Logit (MVMNL) model and Tulabandhula et al. (2023) construct the Bundle Multinomial Logit (BundleMVL-K) model to account for both complementary and substitution effects across categories. Relatedly, Akchen and Caro (2025) analyze size substitution in apparel retail through a choice-based model and show how substitution across adjacent sizes matters for assortment and inventory planning. In empirical research, Mani et al. (2022) use the MNL model to estimate substitution effects in a shopping basket. These papers have made considerable advances in developing the idea that purchasing one product can either increase or decrease the utility of purchasing another product leading to complementary or substitution effects between products, but there has not yet been the development of a large-scale model to directly estimate relationships from data. Thus, these models typically make assumptions regarding the structure of the between-product effects for tractability, such as assuming that these effects are parameterized at the category-level rather than the product-level or that they are symmetric, which can significantly reduce the number of parameters but have not yet been tested empirically.

Another stream of research uses data-driven methods to discover customer purchase patterns in shopping basket data. The foundational work by Agrawal et al. (1993) introduces the concept of mining association rules between product purchases, employing a breadth-first search strategy and candidate-generation to identify itemsets that frequently co-occur in shopping baskets. Manchanda et al. (1999) develop a multivariate probit model of multi-category purchase decisions to distinguish between factors such as coincidence and complementarity in purchases. In more recent papers, text mining and machine learning approaches have been applied to analyze co-purchase shopping behavior. For example, Jacobs et al. (2016) use probabilistic models, including Latent Dirichlet Allocation (LDA) and Mixtures of Dirichlet-Multinomials (MDM), to predict what a customer will purchase next by identifying latent customer topics. Gabel et al. (2019) customize a neural network language model to uncover latent product attributes by analyzing product co-occurrences in shopping baskets. Additionally, Xia et al. (2019) apply a neural-network-based conditional restricted Boltzmann machine to explore cross-product patterns over time in large-scale consumer purchase data. Related work also studies how the shopping environment affects consumer search and purchase behavior. Ngwe et al. (2019) use large-scale field experiments and transaction data to uncover how consumers respond to search frictions in online basket shopping. In another recent paper, Ruiz et al. (2020) propose a random utility model of sequential discrete choice, where customers make purchase decisions one after another and the utility of subsequent purchases depends not only on the attributes of each product but also on all of the previous purchases. Taken together, these approaches use transaction-level basket data to identify co-purchase and sequential patterns for prediction and recommendation, but they do not model how customers form baskets as the result of interacting

product demands. We instead propose a causal product network-based demand model that uses causal structure learning to recover a directed network of effects among individual products, filters out non-causal co-occurrences, and provides an interpretable representation of substitution and complementarity within shopping baskets.

Building on these modeling approaches, a related stream focuses on applications of basket-shopping behavior in operations. Much of this work examines how product co-purchases can be leveraged for inventory management, assortment optimization, pricing, and choice modeling. One stream develops demand and assortment models that incorporate multi-product dependencies to guide inventory and assortment decisions (Agrawal and Smith 2003, Cachon and Kök 2007). A second stream highlights how complementarities, substitution, and cross-category price effects shape purchase incidence, typically through assumptions that interaction patterns operate at the category level rather than at the product level (Chernev 2005, Song and Chintagunta 2006, Lee et al. 2013, Mani et al. 2022). A related set of studies investigates how such purchase patterns differ across brick-and-mortar and online channels, documenting systematic differences in basket variety, similarity, and substitution/complementarity patterns across environments (Wang and Goldfarb 2017, Chintala et al. 2023). We extend this literature by developing a data-driven representation of substitution and complementarity across products, constructed using causal structure learning. In particular, we examine whether basket interactions are more accurately captured at the level of individual products rather than at the category level, and whether the resulting interaction networks differ in structure and sparsity between brick-and-mortar and online channels. Taken together, our findings show that this data-driven representation of substitution and complementarity is both product-specific and channel-specific, with important implications for assortment, pricing, and inventory decisions across retail environments.

To discover the relationships in our model, we introduce a technique known as causal structure learning, which can infer causal relationships from observational data (Meek 2013, Glymour et al. 2019). Causal structure learning has been utilized in the fields of economics and policy-making in the recent years. Examples of this growing literature include uncovering latent macroeconomic models (Hall-Hoffarth 2022), the relationship between climate change and economic growth (Martinoli et al. 2023), inferring causal structures from observational data to validate instrumental variables for the well-known quarter-of-birth and proximity-to-college instruments to estimate the returns to education (Eberhardt et al. 2024), and combining short-term experimental data with long-term observational data to analyze the long-term effects of subsidies on healthy food products (Kaynar and Mitrofanov 2024). In addition to these studies in economics, causal structure learning has been widely applied in other areas, including bioinformatics (e.g., Foraita et al. 2020, Triantafillou et al. 2017), healthcare (e.g., Tu et al. 2019, Shen et al. 2021, Hasan and Gani 2022), and environmental

science (e.g., Ebert-Uphoff and Deng 2014, Runge et al. 2019, Nowack et al. 2020). Our study brings this approach to retail operations by applying causal structure learning to transaction-level shopping-basket data to recover directed product-level interaction networks, assess how well they capture basket-shopping behavior relative to standard category-level specifications, and compare the sparsity and structure of these networks between brick-and-mortar and online channels.

3. Causal Product Network Model and Hypothesis Development

We begin by presenting a causal network model of basket-shopping behavior, then develop hypotheses to evaluate this model against alternative models and to estimate product interaction effects under different assumptions (e.g., category-level versus product-level).

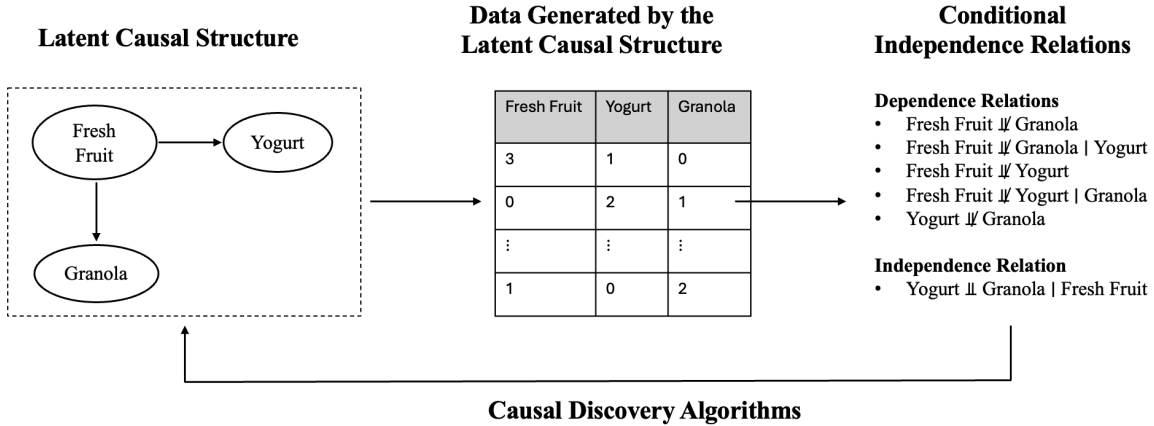
3.1. Causal Product Network Representation of Basket-Shopping Behavior

Suppose that we have basket-shopping data for a homogeneous set of customers. Let $J = \{1, 2, \dots, m\}$ denote the set of product categories, where m is the total number of categories. For each category $j \in J$, let $I_j = \{1, 2, \dots, n_j\}$ be the set of products, where n_j is the number of products in category j . The total number of products across all categories is $N = \sum_{j \in J} n_j$. We define $X_{ijt} \in \mathbb{Z}_{\geq 0}$ as a discrete random variable representing the quantity purchased of product $i \in I_j$ from category j in basket t , where $t = 1, \dots, T$ and T is the number of observed baskets. For notational convenience, we also define $X_{ij} \in \mathbb{Z}_{\geq 0}^{1 \times T}$ as the vector representing the purchase quantities of product i in category j across all T baskets—that is, $X_{ij} = (X_{ij1}, X_{ij2}, \dots, X_{ijT})$. The mapping of products into categories will be useful in our empirical analysis for formulating hypotheses about cross- and within-category causal relationships, but we do not use it as an input to the causal learning method. Indeed, an important aspect of our analysis is that we test the method’s ability to discover relationships without this knowledge.

Each basket is the outcome of purchase decisions made by a customer during the shopping trip. We assume that this purchase behavior of customers is represented by a directed, acyclic graph $\mathcal{G} = (V, E)$, where each product purchase decision is a vertex, i.e., $V = \{X_{ij} \mid i \in I_j, j \in J\}$, and E denotes direct causal relationships from one product purchase to another. Note that, with a slight abuse of notation, we use X_{ij} to denote both the random variable associated with product (i, j) and the corresponding node in the causal graph $\mathcal{G}$, as is common in graphical causal models. We say that an edge from product $X_{i'j'}$ to X_{ij} represents a causal relationship in the sense that the purchase decision of product $X_{i'j'}$ positively or negatively affects the purchase decision of product X_{ij} . Here, $X_{i'j'}$ is said to belong to the parent set of X_{ij} . If products $X_{i'j'}$ and X_{ij} are merely correlated due to other products and do not have a causal relationship, then there is no edge between products $X_{i'j'}$ and X_{ij} . Thus, this approach enables us to distinguish true causal relationships from correlations.

Using the causal graph, we model demand as follows. With each vertex X_{ij} , we associate $\alpha_{ij} \in \mathbb{R}_{\geq 0}$ representing a baseline independent purchase rate for that product, and with each edge $(X_{i'j'}, X_{ij})$, we associate $\beta_{i'j'}^{ij} \in \mathbb{R}$ representing the linear effect derived from each product $X_{i'j'}$ in the parent set $\text{pa}_{\mathcal{G}}(X_{ij})$ to product X_{ij} . The effect $\beta_{i'j'}^{ij}$ can take positive or negative values, representing situations where the demand for one product may cause an increase or a decrease in the demand for another product. The total mean demand rate for X_{ij} is thus given by $\alpha_{ij} + \sum_{X_{i'j'} \in \text{pa}_{\mathcal{G}}(X_{ij})} \beta_{i'j'}^{ij} X_{i'j'}$ and is constrained to be non-negative. Graph $\mathcal{G}$ can be partially connected. For example, if a product is not related to any other product, then it is represented as an isolated vertex with a null parent set. Similarly, if products within a category are substitutes and there is no relationship across categories, then the graph is separable into connected components corresponding to each product category. Thus, this model is flexible in representing all types of causal relationships between pairs of products and the resulting structure of demand. Our goal is to learn the graph $\mathcal{G}$ and the demand parameters from observed data.

Figure 1 Example causal graph and the conditional independence relations it implies. The symbol $\perp\!\!\!\perp$ denotes (conditional) independence, and $\not\perp\!\!\!\perp$ denotes (conditional) dependence.



We illustrate this conceptual framework with an example. Suppose a customer purchases fresh fruit, yogurt, and granola according to the relationships specified by the causal graph shown in Figure 1, i.e., the customer purchases fresh fruit, and with some independent likelihoods, buys yogurt and/or granola with the fresh fruit. Here, fresh fruit is the driver of the purchase and has direct causal effects on yogurt and granola purchases, while there is no direct causal edge between yogurt and granola. As a result, yogurt and granola may appear to be statistically dependent, but once we condition on fresh fruit purchases, the association between yogurt and granola disappears—they become conditionally independent given fresh fruit. If, instead, the purchase of yogurt directly influenced the purchase of granola, the graph would include a direct edge from yogurt to granola and

the conditional independence would no longer hold: yogurt and granola would remain statistically dependent even after conditioning on fresh fruit. Thus, causal graphs are a versatile structure for representing purchase behavior and distinguishing between alternative causal graphs. Due to this, they can have important implications for retail decisions. For example, for the graph presented in Figure 1, a discount on fresh fruit would increase the purchase of yogurt and granola, whereas a discount on yogurt or granola would not increase the purchase of fresh fruit because fresh fruit is the driver product in the graph. As we show later, our hypotheses and empirical analysis reveal such insights into consumer purchasing behavior from the data.

We estimate the CPNs using causal structure learning, which is a novel technique applied to observational data to identify the latent directed graph $\mathcal{G} = (V, E)$ of causal relationships, where V denotes the set of vertices representing the variables in the dataset and E denotes the set of directed edges to be discovered. The goal in this technique is to uncover these underlying causal relationships from purely observational data without the need for randomized experiments. This is possible because causal relationships impose specific constraints on patterns of statistical dependence and independence. Thus, we apply this technique in our setting by mapping product purchase decisions to the vertices and causal relationships between them to the edges of the graph. We discuss the assumptions of this method, the estimation procedure, and identification of the causal model in Section 5.1. Furthermore, Appendix A provides the theoretical background on causal structure learning, illustrative examples, and the complete pseudocode for the PC algorithm, a standard approach for causal learning (Spirtes et al. 2000b).

3.1.1. Alternative directed graphs. The causal product network structure that we propose is sufficiently general to accommodate alternative directed graphs. When the CPN varies across different customer segments or different types of shopping trips, the CPN model can be estimated separately for each segment or trip-type to estimate their respective directed graphs and assess if they are dissimilar from each other. Baskets can be segmented for this purpose by customer demographic data, in-store promotions, type of retail store, and other variables. For instance, in Hypothesis 3, we estimate CPNs separately for brick-and-mortar and online channels.

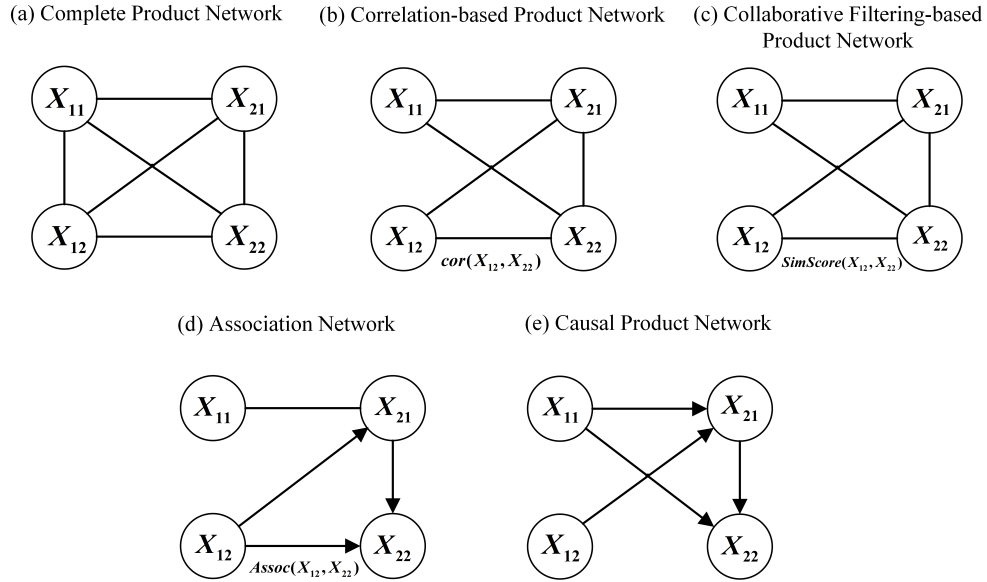
3.2. Hypothesis Development

In this section, we develop our hypotheses to evaluate the ability of CPN to characterize basket-shopping consumer behavior, compare it with alternative theory- and machine learning-based network structures, and utilize it to provide insights.

3.2.1. Comparison of Alternative Network Structures. We compare our approach to several alternative methods for characterizing basket-shopping consumer behavior, drawing from both theoretical foundations and machine learning techniques. Specifically, we consider five different ways to represent interactions among product purchases in a shopping basket:

- (a) **Complete network:** Inspired by Tulabandhula et al. (2023), this model assumes that each product is interconnected with every other product in the basket.
- (b) **Correlation-based network:** Product connections are determined by the Pearson correlation test, capturing the linear association between product purchase patterns.
- (c) **Collaborative filtering-based network:** Product connections are defined using cosine similarity, capturing behavioral similarity based on their co-occurrence in shopping baskets.
- (d) **Association network:** This network is inferred using the procedure by Agrawal et al. (1993), which identifies directed edges based on association rules mined using the Apriori algorithm.
- (e) **Causal network:** This network is inferred using the PC algorithm, which identifies edges based on conditional independence relations among products.

Figure 2 Different types of product interaction networks that can arise from the same purchase data by making different modeling assumptions



Note: Figure 2 illustrates how the same basket-level purchase data can give rise to different product interaction networks depending on the modeling assumption. The complete network assumes full connectivity; the correlation-based network encodes statistically significant correlations; the collaborative filtering-based network reflects similarity in co-occurrence; the association network reflects association rules that exceed both the minimum support and confidence thresholds, as defined by Agrawal et al. (1993); and the causal network, inferred using conditional independence tests, represents directional relationships learned through causal structure learning.

Figure 2 illustrates examples of these networks showing interactions among four products across two categories. We note that all the networks in Figure 2 can be alternative representations obtained from the same real-world shopping basket dataset. For instance, if we assume correlation fully captures relationships between product purchases and use only a correlation-based network, we might incorrectly infer a direct connection between purchases of X_{12} and X_{22} when, in reality, this relationship could be mediated by purchases of X_{21} . Similarly, in the collaborative filtering-based network,

cosine similarity captures only pairwise relationships between product purchases. For the association network, the association rules are based on co-occurrence patterns, and thus, may reflect spurious patterns. Unlike these network structures, the adapted causal structure can remove spurious connections between product purchases and provide directional relations by conducting rigorous conditional independence tests for each relationship. This results in an accurate and parsimonious representation of purchase relationships. Thus, as the first application of our methodology, we compare these network representations on our dataset by testing the following hypothesis.

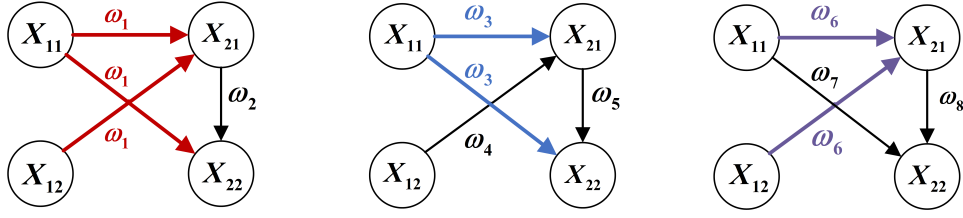
Hypothesis 1 (Network Structures). *A causal product network more accurately represents product purchase relationships in basket shopping compared to complete, correlation-based, collaborative filtering-based, and association networks.*

Our test of this hypothesis is important for modeling basket-shopping behavior because each of these network representations reflects a different set of assumptions about how product purchases relate to each other. The complete network assumes a fully connected structure but relies on regularization techniques to induce sparsity during estimation. The correlation-based, similarity-based, association-based, and causal networks impose structural constraints that limit potential product interactions based on statistical significance, similarity, association rules, or inferred causal relationships. Our statistical comparisons of these models presented in Section 6 show that CPN provides a more accurate representation of basket-shopping behavior than the alternatives.

3.2.2. Specifications for Causal Representations in Basket Shopping. Building on the hypothesis that a causal network more accurately represents basket-shopping behavior, we now investigate various specifications within causal structures. The prior literature shows that there can be different kinds of assumptions regarding consumers’ basket-shopping behavior. At one extreme, Song and Chintagunta (2006) and Jasin et al. (2023) assume that interactions between two products from distinct categories depend solely on the categories to which they belong. At the other extreme, our CPN model allows arbitrary interactions between any pair of products. Other possibilities can lie in between. Figure 3 illustrates these specifications with reference to the causal network shown in Figure 2(e). The products X_{11}, X_{12} and X_{21}, X_{22} belong to categories 1 and 2, respectively. The parameters ω_p for $p = 1, 2, \dots, 8$ correspond to different magnitudes of causal effects. Figures 3(f)-(h) show different conceptual representations of consumer behavior. Specifically, Figure 3(f) considers category-level specification based on Song and Chintagunta (2006) and Jasin et al. (2023), ensuring that the magnitude of causal effects from X_{11} to X_{21} , X_{11} to X_{22} , and X_{12} to X_{21} are identical. Figure 3(g) considers product-to-category specification, ensuring the magnitude of causal effects from X_{11} to X_{21} and X_{11} to X_{22} are identical. Finally, Figure 3(h) shows a category-to-product specification, where we restrict the magnitude of causal effects from X_{11} to X_{21} and X_{12} to X_{21} to be identical.

Figure 3 Alternative causal product network specifications to Figure 2(d) with different types of imposed restrictions

(f) Category-Level Causal Effects (g) Product-to-Category Causal Effects (h) Category-to-Product Causal Effects



The study of these specifications is useful for modeling and estimating consumer shopping behavior because the identification of the correct model has implications for decision-making tasks such as assortment planning, pricing, and promotions. The four models, including Figure 2(e) and Figures 3(f)-(h), exhibit varying levels of generalization: considering causal effects between products represents the most general structure while considering causal effects between categories is the most restricted. These restrictions, if proven accurate, can have critical managerial implications. Additionally, they are also computationally relevant because the number of parameters to be estimated decreases with the restrictions. On the one hand, the more general models allow parameters to be tailored to individual products, but on the other hand, having more parameters might be computationally burdensome and might not provide the best model fit. Thus, balancing this tradeoff, we hypothesize that models capturing product-level interactions provide a more accurate representation of shopping data even when penalizing for model complexity.

Hypothesis 2 (Specification Comparison). *Across the four causal specifications, the product-level causal effects model illustrated in Figure 2(e) most effectively describes the causal relationships in basket shopping (in terms of model fit) compared to alternatives illustrated in Figures 3(f)-(h). In other words, the restrictions imposed by the alternative models do not hold.*

3.2.3. Sparsity of Causal Product Networks Across Channels. The power of our causal network learning approach for archival data is that it can be not only used to construct evidence regarding different model specifications but also applied separately to different retailing contexts to analyze differences in basket-shopping behavior of customers. We now turn our attention to investigating differences in these networks across brick-and-mortar and online channels. Indeed, the prior literature has examined assortment optimization problems across these two channels. For example, Lo and Topaloglu (2022) examine an assortment problem for a physical store aimed at maximizing the total expected revenue of an omnichannel retailer. Gopalakrishnan et al. (2023) compare the impacts of assortment width, defined as the number of unique categories in which a retailer offers products, and assortment depth, defined as the average number of products offered within each category, on order delivery timeliness. Sapra and Kumar (2023) examine the joint

assortment optimization strategy for omnichannel retailers, taking into account three classes of customers who use only the brick-and-mortar channel, only the online channel, or both channels.

The models in the above literature are based on differences in both the cost economics and consumer behavior across the two channels. With regard to empirical evidence for the latter, Chintala et al. (2023) show that there are systematic differences between online and offline grocery shopping. Specifically, online purchases have significantly lower shopping basket variety than brick-and-mortar. However, they do not study causal relationships in consumer purchase behavior and highlight the need for considering causal relationships. Building on this, we hypothesize that the online channel exhibits fewer causal relationships among product purchases within shopping baskets compared to the brick-and-mortar channel (Hypothesis 3). We believe that this difference in sparsity can provide insights for omnichannel assortment optimization by revealing how consumer behavior varies across online and offline environments.

Hypothesis 3 (Sparsity). *Compared to the brick-and-mortar channel, the online channel exhibits fewer causal relationships among product purchases within shopping baskets.*

To test Hypotheses 1 and 2, we construct corresponding structural equation models (SEMs), as discussed in Section 5.2. To test Hypothesis 3, we first construct the causal graphs separately for online and brick-and-mortar baskets using 30 samples. Then, rather than estimating a SEM, we compare the total number of edges in the resulting graphs using a paired t-test.

4. Data Description

We collaborate with Numerator, a prominent market research company known for its first-party, consumer-sourced data. Numerator gathers purchase data from both brick-and-mortar and online channels for a large panel of consumers through multiple sources: retailer loyalty data, mobile app purchases, email receipts, as well as uploaded paper receipts. Their extensive database comprises customer purchase records from more than 17,000 stores, encompassing major retail chains such as Walmart, Costco, and Target, as well as small supermarkets and local grocery stores across the United States. We use their dataset for the year 2021.

Since the number of products in a retail chain is vast, it is not managerially meaningful to construct a single CPN across all the products. Thus, for our analysis, we select five different *themes* for our analysis: each theme is a collection of multiple representative product categories and each category includes several products. For each theme, we consider the most frequently purchased products in each selected department to represent shopping baskets as these products are representative of shopping baskets. Further, to include information about other products within and outside these departments in the product networks, we create aggregated control variables for the total amount of purchases of other products within each selected department and of products outside the selected

departments. These control variables help account for information about other product purchases in shopping baskets, thereby supporting the causal sufficiency assumption in our empirical study. This design of our study provides us with sufficient information to construct and analyze product networks for the five themes and demonstrate the usefulness of this approach. It enables us to conduct five tests of our hypotheses on different themes for a comprehensive evaluation.

Specifically, we focus on the following themes: pasta, quick service restaurant (QSR), bakery, prepared Food, and unprepared food, labeled sequentially as Themes 1 through 5. We chose these themes based on two criteria: (1) items within each theme are available in both brick and mortar and online shopping channels, and (2) the themes are distinct enough to test the hypotheses across a range of scenarios. Theme 1 comprises five departments: pasta & noodle, meat, produce, dairy, and condiments. Theme 2 includes ten departments: QSR beverages, QSR breakfast, QSR sandwiches & wraps, QSR sauces & condiments, QSR Mexican, QSR snack & Sides, QSR desserts, QSR entrees, QSR Italian, and QSR salads. Theme 3 comprises five departments: bakery sweet goods, in-store bakery, packaged bakery, baking & cooking and dairy. Theme 4 includes five departments: frozen foods, canned, deli & prepared foods, beverages, and condiments. Theme 5 comprises five departments: produce, shelf stable meals, meat, herbs & spices, and condiments. Note that product category and department are used synonymously in our data set. The lists of the most frequently purchased product sets for the five themes across both brick-and-mortar and online channels, in the brick-and-mortar channel only, and in the online channel only are shown in Appendix B.

To test Hypotheses 1 and 2, we extract the basket data for each theme including purchases from both brick-and-mortar and online channels. For each theme, we identify the most frequently purchased products within each department in 2021 combined across both channels and choose a total of 50 products. In Themes 1, 3, 4, and 5, which consist of five departments each, we select the top 10 products from each department. In Theme 2, which includes ten departments, we select the top 5 products from each department. After establishing the product set for each theme using the entire dataset, we retrieve all the shopping basket data, including both online and brick-and-mortar channels, for 500 randomly chosen customers who had purchased products from at least three departments within the theme in a single basket. We use these samples as the graph learning datasets for causal structure learning, and collectively refer to them as Data I. Similarly, we extract another set of samples for 500 customers each as the parameter estimation samples for the SEM step and refer to them as Data II. This approach of separate samples for graph learning and parameter estimation avoids overfitting.

To test Hypothesis 3, since the most frequently purchased products differ across the two channels, we consider two product sets for each channel: the most frequently purchased products in brick-and-mortar stores and the most frequently purchased products online. We refer to these as the top brick-and-mortar products and top online products, respectively. For each product set, we retrieve baskets

for each theme in the same manner as above, separating brick-and-mortar and online purchases, termed brick-and-mortar baskets and online baskets. Thus, we create four datasets for each theme as combinations of {brick-and-mortar, online} channels $\times$ {top brick-and-mortar, top online} product sets. We refer to this collection of datasets as Data III, which comprises twenty datasets ($= 4$ scenarios $\times 5$ themes). By considering the most popular products from both brick-and-mortar stores and online platforms and analyzing purchases made through both these channels, we aim to show that the differences between brick-and-mortar purchases and online purchases are consistent across the datasets regardless of the type of products being considered.

Tables 1(a), 1(b), and 1(c) present the statistical summaries for the total number of products purchased per basket for Data I, Data II, and Data III, respectively. As shown in Table 1(a), the median number of purchased items is the same across the five themes, and the mean number is similar, ranging from 2.52 to 3.02. In contrast, the maximum quantity varies more substantially, ranging from 29 to 194 items. Similarly, in Table 1(b), the median number of purchased items remains the same across the five themes, and the mean number is similar, ranging from 2.55 to 2.94. The maximum quantity also varies considerably, ranging from 35 to 236 items. In Table 1(c), the median quantity of purchased items in B&M baskets remains at 2, whereas the median quantity in online baskets varies between 1 and 4. On average, online baskets also contain more items compared to B&M baskets. Additionally, the number of baskets for each theme varies between 5,000 and 15,000 across Data I, Data II, and Data III, with the exception of Theme 2 (Quick service restaurants theme) sample in the online channel. Note that the sample size for Theme 2 in the online channel is smaller, with only 1,776 baskets for the top brick-and-mortar channel products and 2,437 baskets for the top online channel products. We believe this is due to the nature of QSRs, making it attractive for customers to make purchases in person rather than waiting for online orders and delivery.

5. Estimation Method

We estimate the model in two stages to learn the causal structure and estimate effects separately. First, we learn the causal product network $\mathcal{G}$ from transaction-level baskets using the PC algorithm. Section 5.1 summarizes the PC algorithm and states the identifying assumptions required for causal structure learning in our setting. Second, given the learned causal graph, we estimate edge coefficients using a linear structural equation model (SEM), as discussed in Section 5.2.

5.1. Causal Structure Learning and the Underlying Assumptions

In this section, we outline the graph-learning step used to construct the CPN from transaction data. Because within-basket interactions potentially span hundreds of products, exhaustive experimentation is not a scalable way to map cross-item relationships; instead, we use conditional-independence

Table 1 Summary statistics of shopping baskets in Data I, Data II and Data III**(a) Data I (Graph Learning Sample), used for discovering causal DAGs in Hypotheses 1 and 2**

	Total Quantity of Items in Each Basket							
	Min	25%	50%	75%	Max	Mean	SD	N
Theme 1 (Pasta)	1	1	2	4	29	3.02	2.96	14,689
Theme 2 (QSR)	1	1	2	3	105	2.74	2.84	8,858
Theme 3 (Bakery)	1	1	2	3	32	2.52	2.23	10,481
Theme 4 (Prepared Food)	1	1	2	4	61	2.78	2.70	8,345
Theme 5 (Unprepared Food)	1	1	2	3	194	2.62	4.05	14,017

(b) Data II (Parameter Estimation Sample), used for testing Hypotheses 1 and 2

	Total Quantity of Items in Each Basket							
	Min	25%	50%	75%	Max	Mean	SD	N
Theme 1 (Pasta)	1	1	2	4	97	2.94	2.82	14,595
Theme 2 (QSR)	1	1	2	3	35	2.69	2.62	9,408
Theme 3 (Bakery)	1	1	2	3	35	2.55	2.27	10,204
Theme 4 (Prepared Food)	1	1	2	3	99	2.63	2.74	12,530
Theme 5 (Unprepared Food)	1	1	2	3	236	2.67	3.80	12,892

(c) Data III, used for testing Hypothesis 3

<i>Top Brick-and-Mortar Products</i>																
	Brick-and-Mortar Channel								Online Channel							
	Min	25%	50%	75%	Max	Mean	SD	N	Min	25%	50%	75%	Max	Mean	SD	N
Theme 1	1	1	2	4	29	2.92	2.61	14,472	1	2	4	6	27	4.80	3.67	6,664
Theme 2	1	1	2	3	47	2.75	2.57	9,247	1	1	1	2	48	1.91	2.90	1,776
Theme 3	1	1	2	3	41	2.53	2.20	10,114	1	2	3	4	25	3.39	2.39	6,125
Theme 4	1	1	2	3	112	2.64	2.86	11,744	1	1	3	4	60	3.50	3.28	5,719
Theme 5	1	1	2	4	170	3.16	3.77	12,901	1	2	4	6	46	4.81	4.07	5,914
<i>Top Online Products</i>																
	Brick-and-Mortar Channel								Online Channel							
	Min	25%	50%	75%	Max	Mean	SD	N	Min	25%	50%	75%	Max	Mean	SD	N
Theme 1	1	1	2	4	258	2.89	4.10	13,023	1	2	5	8	62	5.78	4.54	6,613
Theme 2	1	1	2	3	44	2.52	2.32	8,159	1	1	1	2	25	1.56	1.37	2,437
Theme 3	1	1	2	3	72	2.72	2.51	9,219	1	2	3	5	24	3.69	2.66	6,502
Theme 4	1	1	2	4	39	2.82	2.89	10,361	1	2	3	5	53	4.02	3.64	6,240
Theme 5	1	1	2	4	165	2.93	3.09	13,056	1	2	4	7	35	5.39	4.42	6,340

Note: (1) Themes 1-5 correspond to pasta, quick service restaurant (QSR), bakery, prepared food, and unprepared food, respectively, where each theme consists of several product categories; (2) N refers to the total number of baskets; (3) SD refers to the standard deviation of product quantity in shopping baskets; (4) Each theme includes 50 focal products and six control products and the above statistics are for the focal products; (5) Quantity of items refers to the total number of products in a basket for the respective theme.

tests to learn a sparse and interpretable dependence structure that guides downstream SEM estimation.

In our study, we use the PC algorithm, one of the most widely used causal structure learning approaches, which utilizes the correspondence between (conditional) independence relations and the underlying graph to iteratively learn causal relations from observational data (Spirtes et al. 2000b). The key idea is to distinguish direct within-basket relationships from associations induced

by other items: the PC algorithm begins with a complete set of candidate links and removes an edge when the two products are found conditionally independent given an appropriate set of other products, thereby filtering out links that can be accounted for by indirect dependence patterns in the basket. Specifically, it operates in two main phases: the first phase discovers the skeleton of the causal structure without specifying the directions of discovered edges and the second phase handles orienting the edges of the skeleton. Because conditional-independence information does not generally identify a unique directed acyclic graph, some edges may remain unoriented, reflecting directional ambiguity given the available data and assumptions. In our implementation, we use the *complete* PC variant, which is order-independent and produces a more informative partially oriented graph while maintaining global consistency. With this variant, we orient an edge only when its direction is implied by the estimated conditional-independence structure and is consistent with the rest of the graph; otherwise, we retain the edge as undirected. Appendix A provides full details, including the pseudo-code of the PC algorithm and an illustrative example.

Causal structure learning relies on three standard assumptions: the causal Markov condition, faithfulness, and causal sufficiency (Spirtes et al. 2000b; see Appendix A for formal definitions). Next, we discuss each assumption and justify its applicability to our empirical setting. (1) The *causal Markov condition* states that a variable is conditionally independent of its non-descendants, given its direct causes (parents). In our setting, this implies that a specific product’s purchase incidence becomes statistically independent of other non-descendant products once we condition on the set of products that directly influence it. This assumption follows naturally from a structural interpretation of the data-generating process, allowing the joint distribution of product purchases to factorize according to the underlying causal graph. This factorization links the causal structure directly to observable conditional independence relations, forming the mathematical basis for our discovery procedure. (2) *Faithfulness* requires that the conditional independencies observed in the data are structural, not accidental. It assumes that all observed independencies are implied by the graph structure, ruling out pathological cases where parameters along multiple causal pathways exactly cancel each other out (e.g., a positive direct effect perfectly offset by a negative indirect effect). Therefore, faithfulness provides a reliable bridge to infer the underlying causal product network from the observed independence structure. (3) The *causal sufficiency* assumption requires that all common causes of any pair of variables in the analyzed set are themselves included in that set. In our study, the object of interest is a causal product network, in which nodes correspond exclusively to products. Accordingly, causal sufficiency is defined with respect to product-level causes. To operationalize this assumption in a setting with a large number of products, we explicitly include the top 50 products within each theme based on purchase frequency in the shopping basket data. In addition, we include control variables that capture the aggregate number of purchases of other

products within the same department and outside the focal theme. These controls capture residual product influences that are not explicitly represented as individual nodes in the network. By jointly modeling focal products and aggregate purchase activity for the remaining products, we mitigate omitted-variable bias arising from unobserved product purchases and approximate causal sufficiency within the product space¹.

Collectively, these assumptions provide the conditions under which the PC algorithm can recover the underlying causal structure of product purchases from observational basket data. When one or more assumptions is violated, the learned graph should be interpreted more cautiously. In that case, the output is best viewed as a data-driven map of directed dependence relations that is consistent with the conditional-independence patterns observed in transactions, rather than as a definitive statement about causal effects. This perspective distinguishes our approach from traditional causal identification strategies that rely on externally generated variation or clearly defined interventions. *Our goal is to use structure learning to extract the dependence “signatures” left in observational purchase data and to leverage them to study basket-shopping behavior in a systematic and scalable way.* Accordingly, when we use the term *causal*, we use it in the specific sense of the causal structure learning framework: the directed structure summarizes which products appear upstream versus downstream in the conditional-dependence patterns of purchases, providing an interpretable representation of how product purchases are organized and interrelated in customer baskets.

Note that the empirical tests of Hypotheses 1–3 in Section 6 are designed to compare how well different network constructions summarize the observed basket data—not to establish the effect of manipulating one product on another, which would require counterfactual (causal) identification. For Hypotheses 1–2, we learn each candidate network on Data I and evaluate the corresponding SEM on an independent sample (Data II) using AIC and likelihood-ratio comparisons; for Hypothesis 3, we learn networks separately in the brick-and-mortar and online channels and compare their sparsity using edge counts across repeated subsamples. Therefore, the empirical support for Hypotheses 1–3 can be read primarily as evidence about relative model performance and systematic cross-channel differences in the dependence structure of basket purchases, regardless of the extent to which the PC assumptions are satisfied in our setting.

¹ We estimate a causal network among product purchase variables. Accordingly, causal sufficiency is interpreted within the product space: we aim to account for common causes that operate through other product purchases. We do not model non-product drivers such as prices, promotions, or stockouts as nodes in the network. Because the data span many retailers and a full calendar year, our results are less likely to be driven by idiosyncratic store- or week-specific shocks. In future work, the product network can be embedded in a structural demand framework that incorporates such non-product drivers.

5.2. Coefficient Estimation

To test our hypotheses, we build the SEMs for corresponding network structures. For Hypothesis 1, we consider five different network structures as shown in Figure 2 in Section 3.2. The complete network representation of product interactions is specified by

$$X_{ij} = \alpha_{ij}^{comp} + \sum_{\substack{i' \in I_j, j' \in J \\ (i', j') \neq (i, j)}} \beta_{i'j'}^{ijcomp} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J. \quad (1)$$

To induce sparsity, reduce variance in the estimated network, and obtain a more interpretable subset of relevant connections, we estimate the model using Lasso and Ridge regularization techniques by adding an ℓ_1 -penalty and an ℓ_2 -penalty, respectively, to the OLS loss function. We refer to the resulting models as the *Lasso-regularized* and *Ridge-regularized complete networks*.

Let cor_{ij} store the purchase of the product significantly correlated with the purchase of the product i of category j (i.e., the p-value ≤ 0.05). The corresponding SEM for the correlation network representation is defined as:

$$X_{ij} = \alpha_{ij}^{cor} + \sum_{X_{i'j'} \in cor_{ij}} \beta_{i'j'}^{ijcor} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J. \quad (2)$$

However, in the presence of sparse basket data, many pairwise correlations may be based on limited co-occurrence and thus subject to high variance and overfitting. To mitigate this, we apply matrix decomposition techniques—specifically, singular value decomposition (SVD) and eigenvalue decomposition (EVD)—to obtain denoised, low-rank approximations of the correlation matrix. These methods are commonly used to reduce noise and reveal the underlying network structure, including correlation structures (Eckart and Young 1936, Alter et al. 2000, Wall et al. 2003). The resulting network structures, derived from the denoised matrices, are known as the *SVD-* and *EVD-decomposed correlation networks*, respectively.

Let sim_{ij} denote the set of product purchases that have high cosine similarity with product X_{ij} , where i indexes the product and j the category. Let $SimScore(X_{ij}, X_{i'j'})$ represent the cosine similarity score between product purchases X_{ij} and $X_{i'j'}$. Following the standard item-item regression-based collaborative filtering approach (Sarwar et al. 2001), we rank all $X_{i'j'}$ by $SimScore(X_{ij}, X_{i'j'})$, and select the top- K most similar products to form the similarity set sim_{ij} . The representation model is defined as follows:

$$X_{ij} = \alpha_{ij}^{sim} + \sum_{X_{i'j'} \in sim_{ij}} \beta_{i'j'}^{ijsim} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J. \quad (3)$$

Let $\mathcal{A}_{ij}$ denote the set of products that have strong association rules directed toward product X_{ij} , where i indexes the product and j the category. Let $Assoc(X_{i'j'}, X_{ij})$ indicate the association rule from product $X_{i'j'}$ to X_{ij} , which is retained if it satisfies the minimum support and confidence

thresholds (Agrawal et al. 1993). The association set $\mathcal{A}_{ij}$ is defined as the collection of all such $X_{i'j'}$ that have valid association rules pointing to X_{ij} . The association network-based representation model is then defined as follows:

$$X_{ij} = \alpha_{ij}^{assoc} + \sum_{X_{i'j'} \in \mathcal{A}_{ij}} \beta_{i'j'}^{ij, assoc} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J. \quad (4)$$

Let $\mathcal{G} = (V, E)$ be the causal graph derived from the PC algorithm and $\text{pa}_{\mathcal{G}}(X_{ij})$ store the parents of the purchase of product i from category j in graph $\mathcal{G}$. The SEM for the causal network representation is defined as:

$$X_{ij} = \alpha_{ij} + \sum_{X_{i'j'} \in \text{pa}_{\mathcal{G}}(X_{ij})} \beta_{i'j'}^{ij} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J. \quad (5)$$

For Hypothesis 2, the three specifications shown in Figure 3 in Section 3.2 are imposed as restrictions on the product-level model presented in Equation (5). We provide the formal equations defining the SEMs for these specifications below.

First, we introduce the category-level specification. Among the existing causal edges in E , let $E_{j'}^j \subseteq E$ store the edges from products in category j' to products in category j , i.e., $E_{j'}^j = \{(X_{i'j'} \rightarrow x_{ij}) \in E, \forall i' \in I_{j'}, i \in I_j\}$. We restrict the coefficients of all the edges in $E_{j'}^j$ to be identical for any $j, j' \in J$. The corresponding SEM for this structure is defined as:

$$X_{ij} = \alpha_{ij} + \sum_{X_{i'j'} \in \text{pa}_{\mathcal{G}}(X_{ij})} \beta_{i'j'}^{ij} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J \quad (6a)$$

$$\text{s.t. } \beta_{i'_1 j'}^{i_1 j} = \beta_{i'_2 j'}^{i_2 j}, \quad i_1, i_2 \in I_j, i'_1, i'_2 \in I_{j'}, j, j' \in J : (X_{i'_1 j'} \rightarrow X_{i_1 j}), (X_{i'_2 j'} \rightarrow X_{i_2 j}) \in E_{j'}^j. \quad (6b)$$

Next, we introduce the product-to-category specification. Let $E_{i'j'}^j \subseteq E$ store the edges from product i' in category j' to any product in category j , i.e., $E_{i'j'}^j = \{(X_{i'j'} \rightarrow X_{ij}) \in E, \forall i \in I_j\}$. We restrict the coefficients of all the edges in $E_{i'j'}^j$ to be identical for any $i' \in I_{j'}, j, j' \in J$. The corresponding SEM for this structure is defined as:

$$X_{ij} = \alpha_{ij} + \sum_{X_{i'j'} \in \text{pa}_{\mathcal{G}}(X_{ij})} \beta_{i'j'}^{ij} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J \quad (7a)$$

$$\text{s.t. } \beta_{i'_1 j'}^{i_1 j} = \beta_{i'_2 j'}^{i_2 j}, \quad i_1, i_2 \in I_j, i'_1, i'_2 \in I_{j'}, j, j' \in J : (X_{i'_1 j'} \rightarrow X_{i_1 j}), (X_{i'_2 j'} \rightarrow X_{i_2 j}) \in E_{i'j'}^j. \quad (7b)$$

Lastly, we introduce the category-to-product specification. Let $E_{j'}^{ij} \subseteq E$ store the edges from any product in category j' to product i from category j , i.e., $E_{j'}^{ij} = \{(X_{i'j'} \rightarrow X_{ij}) \in E, \forall i' \in I_{j'}\}$. We restrict the coefficients of all the edges in $E_{j'}^{ij}$ to be identical for any $i \in I_j, j, j' \in J$. The corresponding SEM for this structure is defined as:

$$X_{ij} = \alpha_{ij} + \sum_{X_{i'j'} \in pa_{ij}} \beta_{i'j'}^{ij} X_{i'j'} + \epsilon_{ij}, \quad \forall i \in I_j, j \in J \quad (8a)$$

$$\text{s.t. } \beta_{i'_1j'}^{ij} = \beta_{i'_2j'}^{ij}, \quad i \in I_j, i'_1, i'_2 \in I_{j'}, j, j' \in J : (X_{i'_1j'} \rightarrow X_{ij}), (X_{i'_2j'} \rightarrow X_{ij}) \in E_{jj'}^{ij}. \quad (8b)$$

These SEMs correspond to different restrictions on the causal effects in Figure 2(d) as follows: Figure 3(e) visualizes the specification in Equation (6) where the causal effects between categories are identical, i.e., $\beta_{11}^{22} = \beta_{12}^{21} = \omega_1$; Figure 3(f) corresponds to Equation (7) where the causal effects from any product to a category are identical, i.e., $\beta_{11}^{21} = \beta_{11}^{22} = \omega_3$; Figure 3(g) corresponds to Equation (8) where the causal effects from any category to a product are identical, i.e., $\beta_{11}^{21} = \beta_{12}^{21} = \omega_6$. Thus, the three specifications are applied as restrictions to the product-level model presented in Equation (5). This enables us to test Hypothesis 2 using the AIC criterion.

When $\mathcal{G}$ is a fully oriented DAG, the system can be triangularized and estimated equation-by-equation without the simultaneity bias. However, causal structure learning methods cannot always determine the direction of every edge, leaving some relationships between product purchases unoriented (see Appendix A for a detailed discussion). To incorporate undirected edges, we define pa_{ij} to include both directed and undirected neighbors, so that any undirected edge enters the equations for both neighboring variables. We note that a challenge in parameter estimation is that such unoriented or bidirectional edges can introduce simultaneity bias when estimating the SEM. The benchmarking models are more prone to this challenge because they often contain numerous bidirectional edges, making identification difficult. In our analysis, we mitigate this issue for causal DAGs and benchmark models through two approaches: (i) by using different samples to construct the causal product network and estimate the SEM, and (ii) applying dimensionality-reduction techniques such as regularization to simplify the benchmark models, thus minimizing the occurrence of bidirectional relationships.

6. Empirical Evidence

In this section, we present the results of our analysis in Sections 6.1 through 6.3, corresponding to Hypotheses 1-3, respectively.

6.1. Tests of Hypotheses 1: Evidence from Causal Product Networks

To test Hypothesis 1, we generate the causal product network, the regularized complete networks, the matrix decomposition-based correlation networks, and the collaborative filtering-based network for each theme. Based on Data I (the graph-learning sample), we first construct the causal product network for each theme. We employ a significance level of 0.05 for the conditional independence tests within the causal discovery procedure. In our study, we use the complete PC algorithm (see

Table 2 Model fit results for CPN compared to alternative network estimation methods in Hypotheses 1

AIC Scores							
	<i>Lasso</i>	<i>Ridge</i>	<i>SVD</i>	<i>EVD</i>	<i>Collaborative Filtering</i>	<i>Association</i>	<i>Causal</i>
Network Structure	(a1)	(a2)	(b1)	(b2)	(c)	(d)	(e)
Theme 1 (Pasta)	567,990	571,658	559,445	559,393	578,273	563,451	557,721
Theme 2 (QSR)	448,529	447,874	437,961	438,198	514,212	441,388	436,085
Theme 3 (Bakery)	324,774	324,075	323,959	308,765	380,065	314,650	304,645
Theme 4 (Prepared Food)	652,517	654,361	647,528	646,568	718,641	647,204	638,588
Theme 5 (Unprepared Food)	928,613	932,847	988,911	928,083	1,063,438	857,323	822,830
Number of Parameters							
Network Structure	(a1)	(a2)	(b1)	(b2)	(c)	(d)	(e)
Theme 1 (Pasta)	762	2,450	498	474	500	157	213
Theme 2 (QSR)	1,424	2,450	714	600	500	109	254
Theme 3 (Bakery)	1,739	2,450	396	420	500	90	203
Theme 4 (Prepared Food)	1,159	2,450	424	364	500	155	176
Theme 5 (Unprepared Food)	246	2,450	262	266	500	92	210

Note: (1) Models a1 and a2 represent the Lasso-regularized complete network and the Ridge-regularized complete network, respectively; models b1 and b2 represent the SVD-decomposed correlation network and the EVD-decomposed correlation network; model c represents the collaborative filtering-based network; model d represents the association network; and model e represents the causal product network; (2) The size of product sets for all the themes is 50; (3) AIC scores are computed by first estimating the networks on Data I, then computing fit statistics for the SEM models on Data II; (4) The likelihood ratio test (LRT) statistics are reported in Appendix C, and all results are statistically significant at the 1% significance level.

Section 5.1 and Appendix A), which applies all applicable rules to orient the edges of the causal skeleton. This algorithm guarantees *order independence*, ensuring the output is invariant to variable processing sequence, and *maximal informativeness*, by orienting the maximum number of edges common to the Markov equivalence class to provide the most specific causal structure supported by the data. In cases where different rules suggest conflicting edge orientations, we retain both directions, resulting in undirected edges. This approach allows for flexibility in representing both directed and undirected relationships among product purchases, making the model more realistic and practically useful. The implementation of the PC algorithm is conducted using the *pcalg* package (Markus Kalisch et al. 2012).

We then generate Lasso-regularized and Ridge-regularized complete product networks for each theme in Data I by regressing each variable on all other variables using Lasso and Ridge regression, which automatically selects important variables through regularization. The regularization parameters (i.e. the strength of the penalty) are selected by cross-validation. In the SVD-decomposed and EVD-decomposed correlation networks, we first construct correlation matrices where edges between nodes are determined by pairwise Pearson correlation tests at the 0.05 significance level. We then apply Singular Value Decomposition (SVD) or Eigenvalue Decomposition (EVD) to the correlation matrix and retain the top components that explain 80% of the variance, yielding a low-rank approximation of the original structure. The simplified matrix is then thresholded to construct a sparse correlation network that reduces noise. For the collaborative filtering-based network, we construct

edges based on item-to-item similarity, computed using cosine similarity between variables representing product purchases. For each product, we follow the top- K rule by connecting it to the 10 most similar products based on cosine similarity scores, and use these connections to define the network structure (Sarwar et al. 2001). For the association network, we follow the procedure of Agrawal et al. (1993) to construct directed edges based on association rules mined using the Apriori algorithm that exceed the minimum support and confidence thresholds. Since the basket shopping data are sparse, we set the minimum support to 0.1% and the minimum confidence to 15%.²

Having obtained all five types of networks using Data I, we formulate and estimate the corresponding SEMs on Data II using the procedure described in Section 3.2.1. As noted in Section 4, we use different samples for these two steps of the analysis to avoid model overfitting.

Table 2 presents the model fit results using Data II (Parameter Estimation Sample) for all the methods. The top half of the table shows comparative analysis using Akaike Information Criterion (AIC) scores, which balances the number of parameters and model fit, and the bottom half provides insights into the number of statistically significant parameters in each model. As shown by Burnham and Anderson (2004), an AIC difference of more than 10 provides strong evidence in favor of the model with the lower AIC. We further supplement the AIC criterion with likelihood ratio tests presented in Appendix C. Note that each estimate is based on an SEM consisting of 50 equations for 50 products. The number of independent variables varies across the specifications and is different in Theme 2 since it consists of 10 categories.

Comparing the results across all the themes in Table 2, we find that the causal product network consistently exhibits the lowest AIC scores. The AIC differences are all much greater than 10, indicating strong empirical evidence that the causal product network significantly outperforms the alternative benchmark models (Burnham and Anderson 2004). Complementing these statistical findings, we conduct likelihood ratio tests (detailed in Appendix C), which confirm the causal product network’s superior performance at the 1% significance level. Therefore, we conclude that the causal network provides a more accurate representation of product relationships in shopping baskets than all the other methods, thereby confirming Hypothesis 1. The matrix decomposition approaches provide the second best performance, and collaborative filtering consistently has the worst performance. Additionally, turning to the bottom half of the table, we observe that the CPN requires significantly fewer parameters compared to all other network structures, except for the association network. For example, in the pasta theme, the Lasso- and Ridge-regularized complete networks require 762 and 2,450 parameters, respectively; the SVD- and EVD-decomposed correlation networks require 498 and 474 parameters; the collaborative filtering-based network requires

² In our study, we find that varying the minimum support between 0.01% and 0.1%, and the minimum confidence between 5% and 15%, yields similar model fit and does not affect the comparative results.

1,970 parameters; whereas the causal product network requires only 213 parameters.³ On the other hand, association network has only 157 edges. This sparsity arises precisely because association networks cannot detect substitution effects—it can account *only* for complementarity effects. In contrast, the causal product network effectively captures both substitution and complementarity while maintaining a parsimonious structure, thus reducing estimation complexity.

The causal product network empirically outperforms alternative network structures because it captures directional, and interpretable relationships among product purchases. Complete and correlation-based networks often result in dense graphs that obscure direct relationships and lead to overfitting. In contrast, the causal product network uses conditional independence tests to isolate direct causal influences, enhancing both interpretability and generalization. Collaborative filtering focuses on co-occurrence patterns and is prone to popularity bias, offering limited insight into the underlying structure of product relationships. Association rule-based networks primarily capture complementary relationships and tend to favor high-demand products, often neglecting substitution effects and underrepresenting interactions involving low-demand items. As a result, they provide a biased and incomplete view of product interactions. The causal product network addresses these limitations by modeling both complementary and substitutive effects within a causal framework. For these reasons, our empirical evidence supports the hypothesis that the causal product network provides a more suitable representation of product relationships.

6.2. Tests of Hypotheses 2: Comparison of Different Specifications

To test Hypothesis 2, we begin our analysis with the final causal graph obtained above. To this graph, we impose constraints for the category-to-category, product-to-category, and category-to-product specifications in the corresponding simultaneous equations models as discussed in §3.2.2. Table 3 presents the model fit results. Comparing the AIC score of (e) with the three alternative causal specifications (f), (g), and (h), we find that the product-level causal model consistently shows the lowest scores. Consistent with our expectations, the differences in AIC scores exceed 10, indicating strong empirical evidence that the product-level product network significantly outperforms the alternative model specifications (Burnham and Anderson 2004). Complementing these statistical findings, we conduct likelihood ratio tests (detailed in Appendix C), which confirms the superior performance of the product-level specification at the 1% significance level. This shows that the product-level specification most effectively captures the causal relationships in basket shopping and the restrictions imposed by the other models do not hold, substantiating Hypothesis 2. This

³ The Ridge-regularized complete network retains all edges between variables because Ridge regression applies an L2 penalty, which shrinks coefficients toward zero but does not set them exactly to zero. As a result, all variables remain in the model with small but non-zero weights, leading to a fully connected network. In contrast, the Lasso-regularized network uses an L1 penalty, which promotes sparsity by shrinking some coefficients exactly to zero.

Table 3 Model fit results for four different specifications of causal product networks Hypotheses 2

AIC Scores				
Network Structure	<i>Product-Level</i> (e)	<i>Category-Level</i> (f)	<i>Product-to-Category</i> (g)	<i>Category-to-Product</i> (h)
Theme 1 (Pasta)	557,721	571,693	569,227	561,288
Theme 2 (QSR)	436,085	452,019	447,819	440,309
Theme 3 (Bakery)	304,645	320,222	316,940	309,806
Theme 4 (Prepared Food)	638,588	647,984	646,745	641,099
Theme 5 (Unprepared Food)	822,830	922,520	915,581	881,728
Number of Parameters				
Network Structure	(e)	(f)	(g)	(h)
Theme 1 (Pasta)	213	33	94	121
Theme 2 (QSR)	254	72	140	156
Theme 3 (Bakery)	203	34	107	122
Theme 4 (Prepared Food)	176	35	93	114
Theme 5 (Unprepared Food)	210	36	90	118

Note: (1) Model e represents the causal product network; models f, g, and h represent the category-level causal effects model, the product-to-category causal effects model, and the category-to-product causal effects model, respectively; (2) The size of product sets for all the themes is 50; (3) Results are computed based on test data (Data II); (4) The likelihood ratio test (LRT) statistics are reported in Appendix C, and all results are statistically significant at the 1% significance level.

result demonstrates that consumers’ basket-shopping behavior is driven by product-level causality rather than category-level effects. Consequently, these findings provide strong empirical support for incorporating granular product-level interactions into choice models of basket-shopping behavior, in both analytical and data-driven research.

As shown in the bottom half of the table, the number of edges in the product-level causal model is slightly larger than in the alternative restriction-based specifications. For example, in the pasta theme, the causal product network includes 213 parameters, compared to 33 for the category-level specification, 94 for the product-to-category specification, and 121 for the category-to-product specification⁴. Moreover, the product-level estimates exhibit substantial *within-category-pair heterogeneity*: effects within the same ordered category block vary widely in magnitude and can even differ in sign (e.g., *Dairy* $\rightarrow$ *Meat* ranges from -0.064 to $+0.369$). Consequently, the equal-effect restrictions imposed by aggregated specifications are unlikely to hold in practice. Furthermore, because aggregation averages over heterogeneous product-level effects within a category, it can mask statistically meaningful signals. Indeed, in the pasta theme we find that 7 of the 33 estimated parameters in the category-level specification are not statistically significant.

We draw two key inferences from these results: (i) the sparsity of the causal product network suggests that the number of significant product-level relationships is manageable, rendering it both

⁴ With five categories in the pasta theme, a category-to-category specification can have at most $5 \times 5 = 25$ ordered category-pair coefficients. In our setting, however, we also include six aggregated control variables as discussed in §4; allowing edges to and from these controls introduces additional ordered blocks (category $\leftarrow$ control and control $\rightarrow$ category), so the number of category-level parameters can exceed 25. The same logic applies to the other themes.

computationally tractable and managerially actionable for retailers to analyze and leverage these dependencies; and (ii) the granularity of these relationships is critical for model fit, suggesting that model misspecification (e.g., through premature aggregation) may materially bias the decision-making tasks derived from choice modeling.

6.3. Test of Hypothesis 3: Brick-and-Mortar Vs Online Basket-Shopping Behavior

To test Hypothesis 3, we build causal product networks separately for each of the 20 datasets in Dataset III across brick-and-mortar and online channels. For each theme and product set, we extract 30 subsamples from the corresponding dataset, each consisting of orders from 100 randomly selected customers. We apply the PC algorithm to discover the causal graph for each sample, and compare the sparsity of the brick-and-mortar and online causal product networks using a paired t-test. Each pair in the test represents the total number of edges in the brick-and-mortar causal network and the corresponding number in the online causal network in the same subsample.⁵

Table 4 Average number of directed edges in DAGs across 30 simulations: Test for Hypothesis 3

	Top B&M Products				Top Online Products			
	B&M	Online	Difference	Num. of Pairs	B&M	Online	Difference	Num. of Pairs
Theme 1 (Pasta)	63.10	55.83	7.27*** (11.10)	30	97.17	81.77	15.40*** (17.05)	30
Theme 2 (QSR)	45.70	36.07	9.63*** (9.55)	30	56.60	35.97	20.63*** (16.94)	30
Theme 3 (Bakery)	68.13	61.83	6.30*** (5.42)	30	64.60	48.57	16.03*** (9.64)	30
Theme 4 (Prepared Food)	12.37	13.93	-1.57 (5.67)	30	13.03	10.50	2.53*** (4.37)	30
Theme 5 (Unprepared Food)	17.23	13.97	3.27*** (4.83)	30	15.87	14.50	1.37* (5.11)	30

Note: (1) * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$; (2) The values for B&M, Online, and Difference correspond to the average number of directed edges across 30 causal graphs discovered from randomly selected subsets for B&M, Online, and the B&M–Online comparison, respectively.

The results of the paired t-test are presented in Table 4. We report the average number of edges across the 30 discovered causal graphs for both the brick-and-mortar and online channels, along with the differences between these two statistics. Our results show that the online channel has a smaller number of edges than the brick-and-mortar channel in 9/10 cases; eight of these are statistically significant at a 0.01 significance level and one at 0.1. For Theme 4, there is a significant difference when top online products are considered, but not when the top brick-and-mortar products are considered. Overall, these results support our hypothesis that online causal product networks are sparser than their brick-and-mortar counterparts. This suggests that online basket purchases exhibit

⁵ Undirected edges and directed edges are counted only once to ensure consistency in our analysis.

a higher degree of independence, with fewer observed complementary and substitution patterns compared to the brick-and-mortar channel.

We next compare the structures of directed graphs obtained for the brick-and-mortar and online channels to obtain further insights into consumer behavior. To present our findings, we focus specifically on the pasta theme (theme 1) within the top brick-and-mortar (B&M) product set. Our analysis of causal edges reveals clear behavioral differences between these two channels. Considering a total of 2,750 edges for 50 products and 6 control variables, we find that 57.4% of edges do not exist in either channel, 12.3% of the edges exist in both channels, and the remaining 30.3% of edges appear exclusively in either the B&M or the online channel. The large difference between 12.3% and 30.3% shows that the two channels have different consumer behavior patterns. Exploring further, we find that edges involving produce items appear substantially more frequently in B&M channel, showing an 86% higher frequency compared to online channel. Specifically, produce-to-condiment edges are notably stronger in B&M channel (132% higher), indicating increased causal connections from fresh produce toward condiments, consistent with meal-planning or recipe-driven shopping. Additionally, edges involving impulse-oriented products, such as lunch packs, are significantly more frequent in B&M channel (108% higher), further highlighting spontaneous purchasing behaviors unique to physical stores. Conversely, online causal edges predominantly involve refrigerated staples such as dairy and processed meats, accounting for approximately 43% and 27% of the online edges, respectively. This pattern reflects routine, convenience-focused replenishment purchasing in the online channel rather than pantry-oriented purchasing. In fact, edges involving shelf-stable pantry categories remain relatively limited online, despite their suitability for convenient online shopping. Collectively, our findings underscore a clear distinction: physical grocery stores are characterized by spontaneous fresh-meal assembly, pantry categories, and impulse purchasing, whereas online grocery shopping predominantly involves regular, convenience-driven replenishment, particularly evident in dairy and processed-meat categories.

Further, the magnitude of spillovers also differs between the B&M and online channels. For each channel, we aggregate all the graphs obtained from the 30 subsamples, considering an edge to exist if it appears at least once in a subsample, and run SEM to estimate the sizes of effects based on this aggregated CPN. We find that the average effect magnitude in the B&M channel is 0.019, which is lower than 0.027 observed in the online channel. Additionally, the standard deviation of effect magnitudes in B&M is 0.028, also lower than 0.048 in the online channel. This indicates that effects in the B&M channel tend to be smaller and less variable compared to those in the online channel even though the number of directed edges tends to be larger as shown in Table 4. Considering all 2,750 edges, 338 (12.3%) exist in both channels with non-zero estimates. Among these, 175 edges (6.4%) have positive effects in both B&M and online channels, while 53 edges (1.9%) have negative

effects in both channels. The remaining 110 edges (4%) exhibit effects with opposite signs across channels.

In addition, we observe that the within-category sign pattern differs by channel. We use the pasta theme as an example. Specifically, there are 9 positive and 16 negative effects in B&M, versus 23 positive and 10 negative effects in the online channel. This suggests that within-category substitution effects for pasta and noodles are stronger when customers shop in physical stores, whereas complementary effects are more prominent when customers shop online. Outside the Pasta & Noodle department, a similar pattern emerges. In the Meat, Produce, Dairy, and Condiments departments, the B&M channel exhibits 22, 46, 26, and 27 positive effects respectively, alongside 13, 24, 24, and 21 negative effects. In contrast, the online channel shows 25, 52, 27, and 22 positive effects and 8, 13, 22, and 13 negative effects in the same departments. These findings further support the notion that substitution effects tend to be stronger in physical stores, while complementary effects dominate in online shopping.

Thus, we conclude that CPNs demonstrate considerable differences in consumer behavior across the brick-and-mortar and online channels. Using these insights, retailers can tailor their recommendation, marketing, and assortment strategies to better align with each channel. For example, assortments and inventories for the online channel should be managed distinctly from those in brick-and-mortar stores. Specifically, online products can be planned with greater independence, as the relative sparsity of the online causal network implies fewer inter-product dependencies. Furthermore, the higher degree of purchase independence online suggests a greater strategic necessity for recommendation engines and cross-promotions to effectively drive basket expansion compared to the brick-and-mortar environment.

7. Managerial Takeaways and Conclusion

Retail demand is realized through shopping baskets, whereas most retail planning is organized around product categories. This paper develops a causal product network (CPN) framework that applies causal structure learning to transaction-level basket data to infer a sparse, directed set of within-basket spillovers, providing an interpretable and scalable way to study cross-product interactions. Our findings have important implications for retailers: category-level planning should account for product-level spillovers that frequently cross category boundaries and differ across shopping channels. Because retailers typically make decisions about assortment, pricing, promotions, shelf space, and inventory within categories, these cross-product links create an organizational mismatch—an action taken in one category can shift demand for specific items outside that category through within-basket interactions. Recognizing these links helps retailers target cross-category coordination to the product pairs that matter most.

Our first finding is that, when we benchmark alternative network structures on the same basket-shopping data, many apparent purchase links reflect spurious patterns rather than meaningful within-basket spillovers. Co-occurrence-based representations therefore tend to produce dense graphs that are difficult to interpret. The CPN applies conditional independence tests to remove spurious correlations between product purchases, retaining only relationships with direct empirical support and estimating signed spillover effects (allowing for both complementarity and substitution). In our benchmarks, the CPN provides the best fit among the representations we consider while remaining parsimonious. For retailers, the implication is that CPNs yield a sparse and interpretable representation of within-basket spillovers that scales to large number of products across various categories. This structure provides a principled basis for prioritizing the cross-category dependencies most relevant for data-driven decision making.

Our second finding is that cross-category spillovers are inherently product-specific. The data do not support restrictions that aggregate interactions to the category level—whether in category-to-category, product-to-category, or category-to-product form. Instead, spillovers vary substantially across product pairs, and they can even differ in sign within the same category pair. As a result, category-pair summaries mask economically important heterogeneity and can misrepresent which cross-category responses are likely to matter. This result makes parsimony more consequential. If category-to-category structure were adequate, cross-category coordination could be organized around a small number of category links. Because it is not, a product-level representation is required. The CPN provides this representation while remaining tractable: it recovers product-level spillovers and yields a sparse directed network over products that concentrates attention on a limited set of links with direct empirical support, both within and across categories.

Finally, our results demonstrate substantial differences in basket-level spillover effects between online and brick-and-mortar channels. The greater sparsity of online CPNs points to a weaker coupling of products within online baskets compared to brick-and-mortar environments. This pattern is consistent with the idea that in-store shopping, through physical exposure, adjacency, and in-store stimuli, facilitates additional within-basket linkages, including impulse additions, whereas online baskets are more list-driven. We also find systematic differences in the content of relationships across channels: in brick-and-mortar settings, links more frequently involve fresh and impulse-oriented items, whereas online links are more concentrated in staple and replenishment products. These results imply that the mechanisms through which retailers pursue basket expansion should be channel-specific. In brick-and-mortar retailing, the denser interaction structure increases the relevance of coordinated merchandising actions tied to in-store exposure, because more spillovers are expressed through the shopping process itself. In online retailing, the sparser interaction structure implies fewer naturally occurring spillovers within baskets and, consistent with our interpretation, a

greater role for deliberate digital mechanisms such as recommendation systems and cross-promotions to surface complements and support basket expansion.

References

- Agrawal N, Smith SA (2003) Optimal retail assortments for substitutable items purchased in sets. *Naval Research Logistics (NRL)* 50(7):793–822, ISSN 0894-069X.
- Agrawal R, Imieliński T, Swami A (1993) Mining association rules between sets of items in large databases. *Proceedings of the 1993 ACM SIGMOD international conference on Management of data*, 207–216.
- Akchen YC, Caro F (2025) On size substitution and its role in assortment and inventory planning. *Manufacturing & Service Operations Management* .
- Alter O, Brown PO, Botstein D (2000) Singular value decomposition for genome-wide expression data processing and modeling. *Proceedings of the National Academy of Sciences* 97(18):10101–10106.
- Aouad A, Feldman J, Segev D, Zhang DJ (2024) The click-based mnl model: A framework for modeling click data in assortment optimization. *Management Science* .
- Aouad A, Segev D (2023) The stability of mnl-based demand under dynamic customer substitution and its algorithmic implications. *Operations Research* 71(4):1216–1249.
- Bai Y, Feldman J, Segev D, Topaloglu H, Wagner L (2023) Assortment optimization under the multi-purchase multinomial logit choice model. *Operations Research* ISSN 0030-364X.
- Burnham KP, Anderson DR (2004) Multimodel inference: understanding aic and bic in model selection. *Sociological methods & research* 33(2):261–304.
- Börsch-Supan A (1990) On the compatibility of nested logit models with utility maximization. *Journal of Econometrics* 43(3):373–388, ISSN 0304-4076.
- Cachon GP, Kök AG (2007) Category management and coordination in retail assortment planning in the presence of basket shopping consumers. *Management Science* 53(6):934–951, ISSN 0025-1909.
- Chernev A (2005) Feature complementarity and assortment in choice. *Journal of Consumer Research* 31(4):748–759, ISSN 1537-5277.
- Chintala SC, Liaukonytė J, Yang N (2023) Browsing the aisles or browsing the app? how online grocery shopping is changing what we buy. *Marketing Science* ISSN 0732-2399.
- Eberhardt F (2017) Introduction to the foundations of causal discovery. *International Journal of Data Science and Analytics* 3:81–91, ISSN 2364-415X.
- Eberhardt F, Kaynar N, Siddiq A (2024) Discovering causal models with optimization: Confounders, cycles, and instrument validity. *Management Science* .
- Ebert-Uphoff I, Deng Y (2014) Causal discovery from spatio-temporal data with applications to climate science. *2014 13th International Conference on Machine Learning and Applications*, 606–613 (IEEE).

-
- Eckart C, Young G (1936) The approximation of one matrix by another of lower rank. *Psychometrika* 1(3):211–218.
- Foraita R, Friemel J, Günther K, Behrens T, Bullerdiek J, Nimzyk R, Ahrens W, Didelez V (2020) Causal discovery of gene regulation with incomplete data. *Journal of the Royal Statistical Society Series A: Statistics in Society* 183(4):1747–1775.
- Gabel S, Guhl D, Klapper D (2019) P2v-map: Mapping market structures for large retail assortments. *Journal of Marketing Research* 56(4):557–580.
- Glymour C, Zhang K, Spirtes P (2019) Review of causal discovery methods based on graphical models. *Frontiers in genetics* 10:524, ISSN 1664-8021.
- Gopalakrishnan S, Matta M, Yourdshahy MI, Choudhary V (2023) Go wide or go deep? assortment strategy and order fulfillment in online retail. *Manufacturing & Service Operations Management* 25(3):846–861.
- Hall-Hoffarth EW (2022) Causal discovery of macroeconomic state-space models .
- Hasan U, Gani MO (2022) Kcrl: A prior knowledge based causal discovery framework with reinforcement learning. *Machine Learning for Healthcare Conference*, 691–714 (PMLR).
- Jacobs BJ, Donkers B, Fok D (2016) Model-based purchase predictions for large assortments. *Marketing Science* 35(3):389–404.
- Jasin S, Lyu C, Najafi S, Zhang H (2023) Assortment optimization with multi-item basket purchase under multivariate mnl model. *Manufacturing & Service Operations Management* ISSN 1523-4614.
- Kaynar N, Mitrofanov D (2024) Long-term policy impact estimation with causal structure learning .
- Lee S, Kim J, Allenby GM (2013) A direct utility model for asymmetric complements. *Marketing Science* 32(3):454–470, ISSN 0732-2399.
- Lo V, Topaloglu H (2022) Omnichannel assortment optimization under the multinomial logit model with a features tree. *Manufacturing & Service Operations Management* 24(2):1220–1240, ISSN 1523-4614.
- Manchanda P, Ansari A, Gupta S (1999) The “shopping basket”: A model for multicategory purchase incidence decisions. *Marketing science* 18(2):95–114.
- Mani V, Thomas DJ, Bansal S (2022) Estimating substitution and basket effects in retail stores: Implications for assortment planning. *Management Science* 68(7):5002–5024.
- Markus Kalisch, Martin Mächler, Diego Colombo, Marloes H Maathuis, Peter Bühlmann (2012) Causal inference using graphical models with the R package pcalg. *Journal of Statistical Software* 47(11):1–26.
- Martinoli M, Moneta A, Pallante G (2023) Calibration and validation of macroeconomic simulation models: A general protocol by causal search. *Available at SSRN 4325509* .
- McFadden D, Train K (2000) Mixed mnl models for discrete response. *Journal of applied Econometrics* 15(5):447–470, ISSN 0883-7252.

- Meek C (2013) Causal inference and causal explanation with background knowledge .
- Ngwe D, Ferreira KJ, Teixeira T (2019) The impact of increasing search frictions on online shopping behavior: Evidence from a field experiment. *Journal of Marketing Research* 56(6):944–959.
- Nowack P, Runge J, Eyring V, Haigh JD (2020) Causal networks for climate model evaluation and constrained projections. *Nature communications* 11(1):1415.
- Pearl J (1988) *Probabilistic reasoning in intelligent systems: networks of plausible inference* (Morgan kaufmann), ISBN 1558604790.
- Pearl J (2009) *Causality* (Cambridge university press).
- Ruiz FJ, Athey S, Blei DM (2020) Shopper. *The Annals of Applied Statistics* 14(1):1–27.
- Runge J, Bathiany S, Bollt E, Camps-Valls G, Coumou D, Deyle E, Glymour C, Kretschmer M, Mahecha MD, Muñoz-Marí J, et al. (2019) Inferring causation from time series in earth system sciences. *Nature communications* 10(1):2553.
- Sapra A, Kumar S (2023) Optimal joint assortment for an omni-channel retailer. *Information Systems Research* ISSN 1047-7047.
- Sarwar B, Karypis G, Konstan J, Riedl J (2001) Item-based collaborative filtering recommendation algorithms. *Proceedings of the 10th international conference on World Wide Web*, 285–295.
- Shen X, Ma S, Vemuri P, Castro MR, Caraballo PJ, Simon GJ (2021) A novel method for causal structure discovery from ehr data and its application to type-2 diabetes mellitus. *Scientific reports* 11(1):21025.
- Song I, Chintagunta PK (2006) Measuring cross-category price effects with aggregate store data. *Management Science* 52(10):1594–1609, ISSN 0025-1909.
- Spirtes P, Glymour C, Scheines R, Kauffman S, Aimale V, Wimberly F (2000a) Constructing bayesian network models of gene expression networks from microarray data .
- Spirtes P, Glymour CN, Scheines R (2000b) *Causation, prediction, and search* (MIT press), ISBN 0262194406.
- Triantafillou S, Lagani V, Heinze-Deml C, Schmidt A, Tegner J, Tsamardinos I (2017) Predicting causal relationships from biological data: Applying automated causal discovery on mass cytometry data of human immune cells. *Scientific reports* 7(1):12724.
- Tu R, Zhang K, Bertilson B, Kjellstrom H, Zhang C (2019) Neuropathic pain diagnosis simulator for causal discovery algorithm evaluation. *Advances in Neural Information Processing Systems* 32.
- Tulabandhula T, Sinha D, Karra SR, Patidar P (2023) Multi-purchase behavior: Modeling, estimation, and optimization. *Manufacturing & Service Operations Management* 25(6):2298–2313, ISSN 1523-4614.
- Verma T, Pearl J (1990) Equivalence and synthesis of causal models. *Proceedings of the Sixth Annual Conference on Uncertainty in Artificial Intelligence*, 255–270.
- Wall ME, Rechtsteiner A, Rocha LM (2003) Singular value decomposition and principal component analysis. *A practical approach to microarray data analysis*, 91–109 (Springer).

- Wang K, Goldfarb A (2017) Can offline stores drive online sales? *Journal of Marketing Research* 54(5):706–719, ISSN 0022-2437.
- Xia F, Chatterjee R, May JH (2019) Using conditional restricted boltzmann machines to model complex consumer shopping patterns. *Marketing Science* 38(4):711–727.

Appendix A: Theoretical Foundations of Causal Structure Learning and the PC Algorithm

Causal structure learning seeks to infer causal relationships from observational data by exploiting patterns of statistical associations and conditional independencies. Inferring causal relationships solely from observational data—without experimental intervention—is inherently challenging. In this section, we review the theoretical foundations of the data-generating mechanism and the key assumptions that enable such inference. Our aim here is to provide a brief overview of the theory behind causal structure learning; for a comprehensive discussion, see Spirtes et al. (2000b) and Eberhardt (2017).

In causal modeling, the latent graph $\mathcal{G} = (V, E)$ represents the underlying structural-equation model whose joint distribution gives rise to the data we observe. Here, V denotes the set of vertices (each corresponding to a random variable), and E denotes the set of directed edges (each encoding a direct causal relationship). In our setting, V represents the set of product purchase variables that we aim to model. Specifically, we define $V = \{X_{ij} \mid i \in I_j, j \in J\}$, where each X_{ij} is a random variable indicating the quantity of product i purchased from category j . Given the causal graph $\mathcal{G} = (V, E)$, the parent set $pa_{\mathcal{G}}(X_{ij})$ of a node $X_{ij} \in V$ includes all products whose purchases directly influence the purchase decision of product X_{ij} . Formally, $pa_{\mathcal{G}}(X_{ij}) = \{X_{i'j'} \mid (X_{i'j'} \rightarrow X_{ij}) \in E\}$.

In practice, the underlying causal graph $\mathcal{G}$ is not directly observed; we only observe basket-level purchase data, which we assume are generated according to $\mathcal{G}$. A key assumption underlying causal structure learning is that each variable is generated as a function of its parents plus an independent noise term. Specifically, we assume a structural equation model (SEM) of the form

$$X_{ij} = f_{ij}(pa_{\mathcal{G}}(X_{ij})) + \epsilon_{ij}, \quad \forall i \in I_j, j \in J,$$

Here, $f_{ij}(\cdot)$ is a deterministic function and ϵ_{ij} is a noise term statistically independent of all other variables and noise terms.⁶

Given the close correspondence between the latent graph $\mathcal{G}$ and the data-generation process, the observed basket-level purchases can be viewed as independent realizations drawn from the structural model over $\{X_{ij}\}$. Causal structure learning methods aim to infer the latent graph $\mathcal{G}$ based on patterns of conditional independence detected in the data. Specifically, by systematically testing conditional independence relations among the observed product purchases, causal structure learning algorithms seek to recover the latent causal structure $\mathcal{G}$ that best explains these observational dependencies.

We define $P_{\mathcal{G}}(V)$ as the joint probability distribution over all product purchases. It specifies how likely different realizations of purchases are under the data-generating process implied by the graph $\mathcal{G}$. Next, we introduce the causal Markov condition, a key assumption that links the structure of $\mathcal{G}$ to the joint distribution $P_{\mathcal{G}}(V)$. Under this condition, the joint distribution factorizes so that each variable is conditionally independent of all its non-descendants given its parent set—thereby specifying the data-generating mechanism and yielding the conditional-independence relations we test to recover the true causal graph.

⁶ Most causal structure learning methods do not require imposing any parametric form on f_{ij} ; for estimation, however, we impose a linear form for estimation in the next section.

Assumption 1 (Causal Markov Condition). *Each variable $X_{ij} \in V$ is conditionally independent of its non-descendants given its parent set $\text{pa}_{\mathcal{G}}(X_{ij})$ in the causal graph $\mathcal{G} = (V, E)$.*

The causal Markov condition, following from the probability distribution definition in terms of the causal structure (Pearl 2009), enables derivation of observed probabilistic independencies from the causal graph.

We next show how graph's structure specifically affects the joint probability distribution. Under the Causal Markov assumption, the joint distribution $P_{\mathcal{G}}(X_{i_1 j_1}, \dots, X_{i_{|V|} j_{|V|}})$ decomposes into the product of each variable's conditional distribution given its parent set. We now derive this graph-driven factorization.

Applying the chain rule to the joint distribution for any permutation of V ,

$$P_{\mathcal{G}}(X_{i_1 j_1}, \dots, X_{i_{|V|} j_{|V|}}) = \prod_{k=1}^{|V|} P_{\mathcal{G}}(X_{i_k j_k} \mid X_{i_1 j_1}, \dots, X_{i_{k-1} j_{k-1}}).$$

Let π be a topological ordering of the nodes in $\mathcal{G}$, so that if $(X_{i' j'} \rightarrow X_{ij}) \in E$ then $\pi(i', j') < \pi(i, j)$. Re-expressing the chain rule in this order gives

$$P_{\mathcal{G}}(X_{i_1 j_1}, \dots, X_{i_{|V|} j_{|V|}}) = \prod_{k=1}^{|V|} P_{\mathcal{G}}(X_{\pi(k)} \mid X_{\pi(1)}, \dots, X_{\pi(k-1)}).$$

By the Causal Markov Condition, each $X_{\pi(k)}$ is independent of all its non-parent predecessors once conditioned on its parent set $\text{pa}_{\mathcal{G}}(X_{\pi(k)})$. Hence every factor simplifies to

$$P_{\mathcal{G}}(X_{\pi(k)} \mid X_{\pi(1)}, \dots, X_{\pi(k-1)}) = P_{\mathcal{G}}(X_{\pi(k)} \mid \text{pa}_{\mathcal{G}}(X_{\pi(k)})),$$

and we obtain the desired graph-driven factorization

$$P_{\mathcal{G}}(X_{i_1 j_1}, \dots, X_{i_{|V|} j_{|V|}}) = \prod_{k=1}^{|V|} P_{\mathcal{G}}(X_{\pi(k)} \mid \text{pa}_{\mathcal{G}}(X_{\pi(k)})).$$

By eliminating the reference to the topological ordering, the factorization can be written as

$$\prod_{(i,j) \in V} P_{\mathcal{G}}(X_{ij} \mid \text{pa}_{\mathcal{G}}(X_{ij})).$$

This factorization establishes a direct link between the graph structure and the observable joint distribution. We then invoke the *faithfulness* assumption, which, together with the Causal Markov assumption, allows us to infer the underlying causal graph from the conditional-independence relations testable in data.

The conditional independence relations observed in the data must exactly match those entailed by the causal graph structure: no spurious independencies may occur, and none implied by the graph may be absent. We formalize this requirement as the faithfulness assumption, which guarantees that the data exhibit precisely the conditional independence patterns specified by the graph. This assumption ensures that observed independencies reflect the underlying causal structure rather than, for example, exact cancellations between causal pathways (Spirtes et al. 2000a).

Assumption 2 (Faithfulness). *The only independencies present in the probability distribution are those implied by the graph structure through the causal Markov condition.*

Lastly, the causal sufficiency assumption ensures that all common causes of any pair of variables in the set V are also contained within V , thereby excluding the existence of any unobserved confounders.

Assumption 3 (Causal Sufficiency). *No variable external to the set V directly affects more than one variable contained within V .*

These assumptions establish a correspondence between the observed data and the underlying graph, enabling causal structure learning algorithms to leverage probabilistic independence relations implied by the data to learn the underlying causal relations. The following theorem formalizes this correspondence:

Theorem 1. (*Spirtes et al. 2000b*). *Let graph $\mathcal{G}$ be the true data generating graph. Under the causal Markov condition, faithfulness and causal sufficiency assumptions, we have:*

- (i) *for all $X_{ij}, X_{i'j'} \in V$, X_{ij} and $X_{i'j'}$ are adjacent in true graph $\mathcal{G}$ if and only if X_{ij} and $X_{i'j'}$ are dependent conditional on any $C \in C_{(ij)(i'j')}$ and*
- (ii) *for all $X_{ij}, X_{i'j'}, X_{i''j''} \in V$ such that X_{ij} is adjacent to $X_{i'j'}$, $X_{i'j'}$ is adjacent to $X_{i''j''}$, and X_{ij} and $X_{i''j''}$ are not adjacent in the true graph $\mathcal{G}$, $X_{ij} \rightarrow X_{i'j'} \leftarrow X_{i''j''}$ is in the true graph $\mathcal{G}$ if and only if X_{ij} and $X_{i''j''}$ are dependent conditional on every set $C \in C_{(ij)(i''j'')}$ such that $X_{i'j'} \in C$,*

where $C_{(ij)(i'j')} = \{C \mid C \subseteq V \setminus \{X_{ij}, X_{i'j'}\}\}$ stores all possible conditioning sets for $X_{ij}, X_{i'j'} \in V$.

Theorem 1 establishes that both the adjacency relations and the orientation of edges in a graph can be inferred from conditional independence relations. The PC algorithm, which is a popular and widely used method in causal structure learning, utilizes this correspondence between (conditional) independence relations and the underlying graph to iteratively learn the underlying causal relations from observational data (Spirtes et al. 2000b). Specifically, the PC algorithm uses the principles outlined in Theorem 1 and operates in two main phases: the first phase discovers the skeleton of the causal structure without specifying the directions of discovered edges and the second phase handles orienting the edges of the skeleton. The pseudo-codes for these two phases of the PC algorithm are provided in Algorithms A1 and A2, respectively.

In Algorithm A1, we start with a complete undirected graph in which each pair X_{ij} and $X_{i'j'}$ in V is connected by an undirected edge. We then evaluate each pair of variables, X_{ij} and $X_{i'j'}$, for probabilistic independence by considering subsets C in $C_{(ij)(i'j')} = \{C \mid C \subseteq V \setminus \{X_{ij}, X_{i'j'}\}\}$, i.e., all subsets of V excluding the focal pair of variables. The size of each subset, denoted by $|C|$, incrementally increases throughout the testing process. If X_{ij} and $X_{i'j'}$ are found to be independent given a subset C , we remove the edge between them, following part (i) of Theorem 1. The output of Algorithm A1 is the skeleton of the causal structure, which represents the potential causal relations without directionality among the variables and it serves as the input for the next phase of the algorithm. Algorithm A2 focuses on orienting the edges within the graph following part (ii) of Theorem 1. We proceed by examining conditional independence relations between variable triplets $X_{ij}, X_{i'j'}, X_{i''j''} \in V$ such that X_{ij} is adjacent to $X_{i'j'}$ and $X_{i'j'}$ is adjacent to $X_{i''j''}$ and X_{ij} and $X_{i''j''}$ are not adjacent in the skeleton returned by Algorithm A1. If X_{ij} and $X_{i''j''}$ are probabilistically dependent with respect to conditioning set $C = \{X_{i'j'}\}$ then we orient $X_{ij}-X_{i'j'}-X_{i''j''}$ as $X_{ij} \rightarrow X_{i'j'} \leftarrow X_{i''j''}$. The algorithm then employs Meek’s rules for further edge orientation, ensuring that all the orientations are consistent with Theorem 1 and the resulting graph does not include cycles (Meek 2013). The PC algorithm offers flexibility in selecting independence tests appropriate for the specific domain and enables the choice of preferred methods for correcting multiple hypothesis testing. These choices depend on factors such as sample

Algorithm A1: PC algorithm - Skeleton Discovery (Spirtes et al. 2000b)

Input: I_j , J and $V = \{X_{ij} \text{ for } i \in I_j, j \in J\}$.

Output: $\mathcal{G} = (V, E)$.

Initialization: $\mathcal{G} = (V, E)$ where $E = \{(X_{ij} - X_{i'j'}), \forall X_{ij}, X_{i'j'} \in V \text{ such that } X_{ij} \neq X_{i'j'}\}$, $\mathcal{N}_{\mathcal{G}}(X_{ij}) = \{X_{i'j'} \in V \setminus X_{ij}\}$.

1. Initialize $\ell = 0$.
 2. **for** an ordered pair $X_{ij}, X_{i'j'} \in V$ where $X_{i'j'} \in \mathcal{N}_{\mathcal{G}}(X_{ij})$ and $|\mathcal{N}_{\mathcal{G}}(X_{ij}) \setminus X_{i'j'}| \geq l$:
 3. **for** $C \in \mathcal{C}_{(ij)(i'j')}$ with $|C| = \ell$:
 4. **if** $X_{ij} \perp\!\!\!\perp X_{i'j'} \mid C$:
 5. Remove the edge $(X_{ij} - X_{i'j'})$ from E .
 6. Update $\mathcal{N}_{\mathcal{G}}(X_{ij}) = \mathcal{N}_{\mathcal{G}}(X_{ij}) \setminus X_{i'j'}$ and $\mathcal{N}_{\mathcal{G}}(X_{i'j'}) = \mathcal{N}_{\mathcal{G}}(X_{i'j'}) \setminus X_{ij}$.
 7. **Break**.
 8. **if** $\ell < |V| - 2$:
 9. Update $\ell = \ell + 1$. Go to Step 2.
 10. **else** $\ell = |V| - 2$:
 11. **Break**.
 12. Return $\mathcal{G} = (V, E)$.
-

Algorithm A2: PC algorithm - Orienting Edges (Spirtes et al. 2000b)

Input: $\mathcal{G} = (V, E)$.

Output: $\mathcal{G} = (V, E)$.

1. **for** $X_{ij}, X_{i'j'}, X_{i''j''} \in V$ where $\{(X_{ij} - X_{i'j'}), (X_{i'j'} - X_{i''j''})\} \subseteq E$ & $\{(X_{ij} - X_{i''j''})\} \not\subseteq E$:
 2. **if** $X_{ij} \not\perp\!\!\!\perp X_{i''j''} \mid X_{i'j'}$:
 3. Orient $X_{ij} - X_{i'j'} - X_{i''j''}$ as $X_{ij} \rightarrow X_{i'j'} \leftarrow X_{i''j''}$.
 4. Update $E = (E \setminus \{(X_{ij} - X_{i'j'}), (X_{i'j'} - X_{i''j''})\}) \cup \{(X_{ij} \rightarrow X_{i'j'}), (X_{i'j'} \leftarrow X_{i''j''})\}$.
- Meek rules (See Figure 5 for illustration):**
4. **(R1)** **for** $X_{ij}, X_{i'j'}, X_{i''j''} \in V$ where $X_{ij} \notin \mathcal{N}_{\mathcal{G}}(X_{i''j''})$ and $\{(X_{ij} \rightarrow X_{i'j'}), (X_{i'j'} - X_{i''j''})\} \subseteq E$:
 5. Update $E \leftarrow (E \cup \{(X_{i'j'} \rightarrow X_{i''j''})\}) \setminus \{(X_{i'j'} - X_{i''j''})\}$.
 6. **(R2)** **for** $X_{ij}, X_{i'j'}, X_{i''j''} \in V$ where $\{(X_{ij} \rightarrow X_{i'j'}), (X_{i'j'} \rightarrow X_{i''j''}), (X_{ij} - X_{i''j''})\} \subseteq E$:
 7. Update $E \leftarrow (E \cup \{(X_{ij} \rightarrow X_{i''j''})\}) \setminus \{(X_{ij} - X_{i''j''})\}$.
 8. **(R3)** **for** $X_{ij}, X_{i'j'}, X_{i''j''}, x_{i'''j'''} \in V$ where $X_{i'j'} \notin \mathcal{N}_{\mathcal{G}}(x_{i'''j'''})$ &
 9. $\{(X_{ij} - X_{i'j'}), (X_{i'j'} \rightarrow X_{i''j''}), (X_{ij} - x_{i'''j'''})\} \subseteq E$:
 10. Update $E \leftarrow (E \cup \{(X_{ij} \rightarrow X_{i''j''})\}) \setminus \{(X_{ij} - X_{i''j''})\}$.
 11. **(R4)** **for** $X_{ij}, X_{i'j'}, X_{i''j''}, x_{i'''j'''} \in V$ where $X_{ij} \notin \mathcal{N}_{\mathcal{G}}(X_{i''j''})$ &
 12. $\{(X_{ij} \rightarrow X_{i'j'}), (X_{i'j'} \rightarrow X_{i''j''}), (X_{ij} - x_{i'''j'''})\} \subseteq E$:
 13. Update $E \leftarrow (E \cup \{(x_{i'''j'''} \rightarrow X_{i''j''})\}) \setminus \{(x_{i'''j'''} - X_{i''j''})\}$.
 14. **Return** $\mathcal{G} = (V, E)$ where $E = E$.
-

size, the number of variables, whether the variables are categorical or continuous, and the assumptions made about the parametric nature of the causal relationships.

In general, the independence structure seen in observational data is not guaranteed to uniquely identify the underlying causal graph. Two graphs with different structures are said to be *Markov equivalent* if they have the same independence structure (Verma and Pearl 1990). In the context of causal structure learning, the Markov equivalence class of the true, data-generating graph is the limit of what can be learned about the causal structure from the independence structure in the data. The PC algorithm has been shown to be asymptotically correct, meaning in the large-sample limit it discovers the true data-generating graph up to

an equivalence class, given Assumptions 1-3 (Spirtes et al. 2000b). This implies that the edge set E returned by Algorithm A1 represents a Markov equivalence class of graphs and it might include undirected edges. In the context of our study, having $(X_{i'j'} \rightarrow X_{ij}) \in E$ implies purchasing product i' from category j' has a *direct* causal impact on purchasing product i from category j . Whereas having $(X_{i'j'} - X_{ij}) \in E$ implies an unspecified causal relationship between the products. In other words, $(X_{i'j'} - X_{ij})$ suggests that there is an edge between $X_{i'j'}$ and X_{ij} , but it could point in either direction.

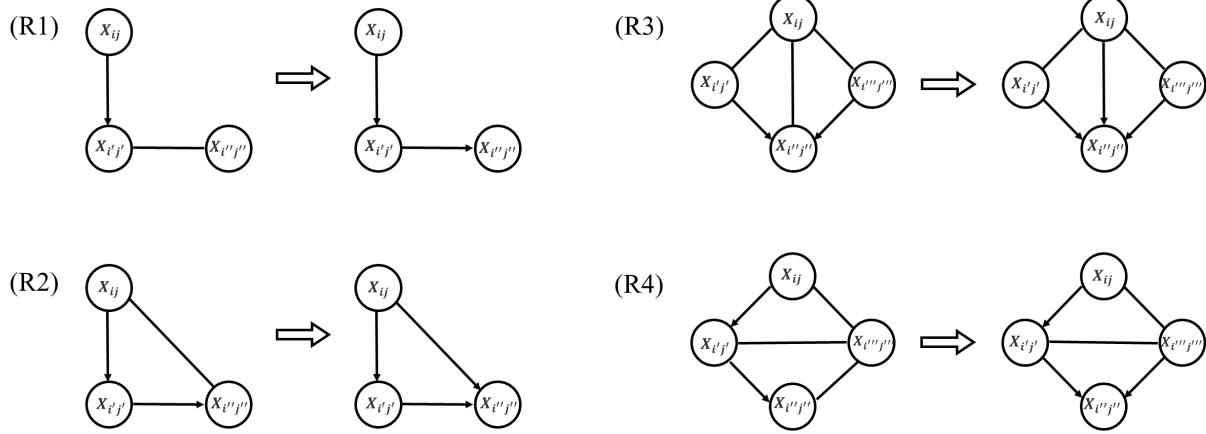
Figure 4 illustrates the steps of the PC algorithm for an example. For a given true data-generating graph over product purchases x_{11} and x_{21} in category 1, and x_{12} and x_{22} in category 2, the PC algorithm conducts a series of conditional independence tests using the data generated accordingly with this graph. Algorithm A1 starts with testing for marginal independencies ($\ell = 0$). Since x_{11} and x_{12} are marginally independent, denoted as $x_{11} \perp\!\!\!\perp x_{12}$, the edge between x_{11} and x_{12} is removed. Then, we start testing for conditional independencies. At the first order ($\ell = 1$), since $x_{11} \perp\!\!\!\perp x_{22} \mid x_{21}$ and $x_{12} \perp\!\!\!\perp x_{22} \mid x_{21}$, the edges between x_{11} and x_{22} and x_{12} and x_{22} are removed. Since no further independencies are found at the second order ($\ell = 2$), we obtain the skeleton of the graph. Using this skeleton, Algorithm A2 orients the edges using the dependency relations. Since $x_{11} \not\perp\!\!\!\perp x_{12} \mid x_{21}$, we must have $x_{11} \rightarrow x_{21} \leftarrow x_{12}$ following part (ii) of Theorem 1. Finally, we orient the remaining edges using Meek's rules, resulting in $x_{21} \rightarrow x_{22}$, forming the directed acyclic graph that captures the causal relationships. In this example, we are able to identify a unique graph, indicating that there is only one graph in the Markov equivalence class of the underlying graph.⁷

Figure 4 Illustration of the PC algorithm steps for an example

	Algorithm A1: PC algorithm - Skeleton Discovery			Algorithm A2: PC algorithm - Orienting Edges	
	$\ell = 0$: Zero-th Order Independencies	$\ell = 1$: First Order Independencies	$\ell = 2$: Second Order Independencies	V - Structure Discovery	Orient Remaining Edges
True Graph Complete Undirected Graph	$x_{11} \perp\!\!\!\perp x_{12}$	$x_{11} \perp\!\!\!\perp x_{22} \mid x_{21}$ $x_{12} \perp\!\!\!\perp x_{22} \mid x_{21}$	None	$x_{11} \not\perp\!\!\!\perp x_{12} \mid x_{21}$	x_{21} cannot be the collider between x_{11} and x_{22}

Furthermore, Figure 5 graphically illustrates the four Meek rules that correspond to (R1), (R2), (R3), and (R4) in Algorithm A2.

⁷ As previously mentioned, unique identification of the causal graphs is not always guaranteed.

Figure 5 Meek rules

Appendix B: Product Lists of Five Themes

Tables 5, 6, and 7 display the most frequently purchased products across both the brick-and-mortar and online channels, the brick-and-mortar channel only, and the online channel only, respectively. Due to Numerator's data disclosure policy, we have concealed the brand information and are only displaying the product lists without brand details.

(See next page for the tables.)

Table 5 Top B&M and online products

Theme 1 (Pasta)	Pasta & Noodle	Spaghetti, Dry, Macaroni, Penne, Rotini
	Meat	Steaks, Ground Beef, Chicken, Lunch Packs, Whole Cuts & Roasts, Pork Chop, Sausage-Pork
	Produce	Bananas, Onions, Tomatoes, Apples, Grapes, Avocado, Cucumber, Potatoes, Lettuce, Strawberries
	Dairy	Milk, Coffee Creamers, Shredded Cheese, Cream Cheese, Sour Cream, Greek Yogurt
	Condiments	Nut Butters, Salad Dressings, Salsa, Hummus, Mayonnaise, Jam, Jelly & Marmalades
Theme 2 (QSR)	QSR Beverages	Cola, Pepper-Style Soda, Hot Coffee, Citrus & Berry Soda
	QSR Breakfast	Hash Browns, Donuts, Sausage Breakfast Sandwich, Ham Breakfast Sandwich
	QSR Sandwiches & Wraps	Beef Burger, Crispy Chicken Sandwich
	QSR Sauces & Condiments	Ketchup, BBQ Sauce, Mild Sauce, Ranch Dip, Hot Sauce
	QSR Mexican	Chicken Quesadilla, Steak Taco, Veggie Burrito, Steak Nachos, Steak Burrito
	QSR Snack & Sides	French Fries, Potato Chips
	QSR Desserts	Milk Shake, Pie, Cookies, Ice Cream Cone
	QSR Entrees	Chicken Nuggets, Bone-In Chicken
	QSR Italian	Hand Tossed Pizza, Pan Pizza, Thin Crust Pizza, Breadsticks
	QSR Salads	Chicken Cobb Salad, Chicken Garden Salad, Southwest Chicken Salad
Theme 3 (Bakery)	Bakery Sweet Goods	Donuts, Muffins, Cakes, Cookies, Snack Pies, Pastries, Pies
	In-Store Bakery	Bread & Breadsticks, Rolls, Bagels, Croissants, Tortillas, Italian Bread, Buns
	Packaged Bakery	Tortillas, Bagels, Buns, English Muffins, Rolls, Wheat Bread, White Bread
	Baking & Cooking	Pasta & Pizza Sauces, Sugar, Barbecue Sauce, Baking Chips, Pudding, Custard & Mousse Mix, Marshmallows, Tomato Sauce, Paste & Puree, Gelatin & Jello Mix, Cake Mixes, Dessert Syrups
	Dairy	Milk, Coffee Creamers, Shredded Cheese, Cream Cheese, Sour Cream, Greek Yogurt
Theme 4 (Prepared Food)	Frozen Foods	Frozen Breakfast, Frozen Chicken, French Fries, Single Serve Meals, Packaged Ice Cream, Pizza Bites/Rolls
	Canned	Prepared Beans, Canned Tuna, Canned Tomatoes, Variety Beans, Applesauce, Fruit Cups, Canned Green Beans
	Deli & Prepared Foods	Chicken-Prepared, Deli Salad, Pork-Deli, Ready-Made Sandwiches & Wraps-Prepared, Turkey-Deli, Sushi-Prepared
	Beverages	Cola, Sports Drinks, Citrus & Berry Soda, Still Water, Seltzers & Sparkling Water
	Condiments	Nut Butters, Salad Dressings, Salsa, Hummus, Mayonnaise, Jam, Jelly & Marmalades
Theme 5 (Unprepared Food)	Produce	Bananas, Onions, Tomatoes, Apples, Grapes, Avocado, Cucumber, Potatoes, Lettuce, Strawberries
	Shelf Stable Meals	Canned Soups, Mac & Cheese, Ramen & Noodle Soups, Potato Mixes, Pasta Dishes, Rice Dishes
	Meat	Steaks, Ground Beef, Chicken, Lunch Packs, Whole Cuts & Roasts, Pork Chop, Sausage, Pork-Mainstream
	Herbs & Spices	Mexican Seasoning, Grill Seasoning, Garlic Powder & Garlic Salt, Chili Seasoning
	Condiments	Nut Butters, Salad Dressings, Salsa, Hummus, Mayonnaise, Jam, Jelly & Marmalades

Note: (1) This table displays product names without brand information, and duplicate names across different brands have been removed; (2) The top products are those that appear most frequently in both online and brick-and-mortar baskets.

Table 6 Top B&M products

Theme 1 (Pasta)	Pasta & Noodle	Spaghetti, Penne, Dry, Rotini, Macaroni
	Meat	Chicken, Lunch Packs, Pork-Mainstream, Steaks, Ground Beef, Pork Chop, Sausage-Pork, Whole Cuts & Roasts
	Produce	Cucumber, Apples, Onions, Tomatoes, Potatoes, Bananas, Strawberries, Grapes, Avocado, Lettuce
	Dairy	Greek Yogurt, Shredded Cheese, Coffee Creamers, Milk, Sour Cream, Cream Cheese
	Condiments	Salsa, Nut Butters, Jam, Jelly & Marmalades, Hummus, Mayonnaise, Salad Dressings
Theme 2 (QSR)	QSR Beverages	Citrus & Berry Soda, Pepper-Style Soda, Frappuccino, Cola
	QSR Breakfast	Sausage Breakfast Sandwich, Hash Browns, Donuts, Ham Breakfast Sandwich
	QSR Sandwiches & Wraps	Beef Burger, Crispy Chicken Sandwich
	QSR Sauces & Condiments	Ranch Dip, Mild Sauce, BBQ Sauce, Ketchup, Hot Sauce
	QSR Mexican	Steak Taco, Veggie Burrito, Chicken Quesadilla, Steak Nachos, Chicken Taco
	QSR Snack & Sides	French Fries
	QSR Desserts	Pie, Milk Shake, Cookies
	QSR Entrees	Bone-In Chicken, Chicken Nuggets
	QSR Italian	Breadsticks, Pan Pizza, Thin Crust Pizza, Hand Tossed Pizza
	QSR Salads	Chicken Garden Salad, Chicken Cobb Salad, Southwest Chicken Salad
Theme 3 (Bakery)	Bakery Sweet Goods	Cookies, Packaged Muffins, Snack Pies, Pastries, Pies, Muffins, Donuts, Cakes
	In-Store Bakery	Tortillas, Italian Bread, Buns, Croissants, Bread & Breadsticks, Rolls, Bagels
	Packaged Bakery	Bagels, Buns, Wheat Bread, Sandwich Bread, White Bread, Tortillas, English Muffins, Rolls
	Baking & Cooking	Sugar, Cake Mixes, Gelatin & Jello Mix, Barbecue Sauce, Dessert Syrups, Pudding, Custard & Mousse Mix, Baking Chips, Marshmallows, Pasta & Pizza Sauces, Tomato Sauce, Paste & Puree
	Dairy	Greek Yogurt, Shredded Cheese, Coffee Creamers, Milk, Sour Cream, Cream Cheese
Theme 4 (Prepared Food)	Frozen Foods	French Fries, Syrup Carriers-Frozen Breakfast, All Other Single Serve Meals, Packaged Ice Cream, Nutrition Single Serve Meals, Frozen Chicken, Prepared Entrees-Frozen Breakfast
	Canned	Fruit Cups, Canned Tomatoes, Applesauce, Canned Green Beans, Canned Tuna, Variety Beans, Prepared Beans
	Deli & Prepared Foods	Turkey-Deli, Ready-Made Sandwiches & Wraps-Prepared, Pork-Deli, Chicken-Prepared, Deli Salad, Sushi-Prepared
	Beverages	Still Water, Seltzers & Sparkling Water, Pepper & Skipper Soda, Sports Drinks, Citrus & Berry Soda, Cola
	Condiments	Salsa, Nut Butters, Jam, Jelly & Marmalades, Hummus, Mayonnaise, Salad Dressings
	Produce	Cucumber, Apples, Onions, Tomatoes, Potatoes, Bananas, Strawberries, Grapes, Avocado, Lettuce
Theme 5 (Unprepared Food)	Shelf Stable Meals	Ramen & Noodle Soups, Pasta Dishes, Canned Soups, Rice Dishes, Potato Mixes, Mac & Cheese
	Meat	Chicken, Lunch Packs, Pork-Mainstream, Steaks, Ground Beef, Pork Chop, Sausage-Pork, Whole Cuts & Roasts
	Herbs & Spices	Chili Seasoning, Garlic Powder & Garlic Salt, Mexican Seasoning, Grill Seasoning
	Condiments	Salsa, Nut Butters, Jam, Jelly & Marmalades, Hummus, Mayonnaise, Salad Dressings

Note: (1) This table displays product names without brand information, and duplicate names across different brands have been removed; (2) The top products are those that appear most frequently in brick-and-mortar baskets.

Table 7 Top online products

Theme 1 (Pasta)	Pasta & Noodle	Spaghetti, Penne, Dry, Rotini, Macaroni, Lasagna, Bow-Tie
	Meat	Chicken, Lunch Packs, Pork-Mainstream, Pork Bacon, Steaks, Ground Beef, Sausage-Pork
	Produce	Cucumber, Salad Greens, Apples, Onions, Tomatoes, Bananas, Grapes, Avocado, Lettuce
	Dairy	Cream, Butter, Shredded Cheese, Natural Sliced Cheese, Coffee Creamers, Milk, Cream Cheese
	Condiments	Salsa, Nut Butters, Jam, Jelly & Marmalades, Hummus, Mayonnaise, Salad Dressings
Theme 2 (QSR)	QSR Beverages	Cola, Hot Coffee, Lemonade
	QSR Breakfast	Bagels, Bakery Other, Hash Browns, Muffins
	QSR Sandwiches & Wraps	Grilled Chicken Sandwich, Crispy Chicken Sandwich, Bacon Breakfast Sandwich
	QSR Sauces & Condiments	Hot Sauce, Ketchup, Mild Sauce, Beef Burger, Jam & Jelly
	QSR Mexican	Chicken Quesadilla, Steak Taco
	QSR Snack & Sides	French Fries
	QSR Desserts	Cookies, Milk Shake, Pie
	QSR Entrees	Chicken Nuggets, Bone-In Chicken
	QSR Italian	Hand Tossed Pizza, Breadsticks, Cheese Bread, Pan Pizza
	QSR Salads	Chicken Garden Salad, Chicken Cobb Salad, Southwest Chicken Salad
Theme 3 (Bakery)	Bakery Sweet Goods	Cookies, Packaged Muffins, Snack Pies, Muffins, Donuts, Pies
	In-Store Bakery	Tortillas, Buns, Croissants, Bread & Breadsticks, Rolls, Ciabatta
	Packaged Bakery	Bagels, Buns, Wheat Bread, White Bread, Tortillas, English Muffins, Rolls
	Baking & Cooking	Distilled Vinegar, Sugar, Barbecue Sauce, Pudding, Custard & Mousse Mix, Baking Chips, Marshmallows, Pasta & Pizza Sauces, Tomato Sauce, Paste & Puree
	Dairy	Cream, Butter, Shredded Cheese, Natural Sliced Cheese, Coffee Creamers, Milk, Cream Cheese
Theme 4 (Prepared Food)	Frozen Foods	Frozen Mixed Vegetables, French Fries, Syrup Carriers-Frozen Breakfast, Frozen Broccoli, Packaged Ice Cream, Nutrition Single Serve Meals, Frozen Chicken, Prepared Entrees-Frozen Breakfast
	Canned	Fruit Cups, Canned Tomatoes, Applesauce, Canned Green Beans, Canned Tuna, Canned Corn, Variety Beans, Prepared Beans
	Deli & Prepared Foods	Turkey-Deli, American Cheese (Deli), Ready-Made Sandwiches & Wraps-Prepared, Pork-Deli, Deli Salad
	Beverages	Still Water, Seltzers & Sparkling Water, Fruit Juice, Sports Drinks, Cola
	Condiments	Salsa, Nut Butters, Jam, Jelly & Marmalades, Hummus, Mayonnaise, Salad Dressings
Theme 5 (Unprepared Food)	Produce	Fresh Cucumber, Salad Greens, Apples, Onions, Tomatoes, Bananas, Grapes, Avocado, Lettuce
	Shelf Stable Meals	Ramen & Noodle Soups, Pasta Dishes, Canned Soups, Rice Dishes, Mac & Cheese
	Meat	Chicken, Lunch Packs, Pork-Mainstream, Pork Bacon, Steaks, Ground Beef, Sausage-Pork
	Herbs & Spices	Chili Seasoning, Onion Powder, Grill Seasoning, Cinnamon (Powder & Sticks), Mexican Seasoning, Garlic Powder & Garlic Salt
	Condiments	Salsa, Nut Butters, Jam, Jelly & Marmalades, Hummus, Mayonnaise, Salad Dressings

Note: (1) This table displays product names without brand information, and duplicate names across different brands have been removed; (2) The top products are those that appear most frequently in online baskets.

Appendix C: Statistical Significance of Model Comparisons

In this paper, we evaluate model fit using the Akaike Information Criterion (AIC) when testing Hypotheses 1 and 2. Alternatively, we can apply the Likelihood Ratio Test (LRT), which, under standard regularity conditions, yields a test statistic that follows a chi-square distribution under the null hypothesis (Burnham and Anderson 2004). Suppose we have two models:

- Model $\mathcal{M}_0$: a causal network model (model e in Sections 3.2.1)
- Model $\mathcal{M}_1$: the network structure with the lowest AIC score among the alternative structures or causal representation specifications.

$\mathcal{M}_0$ constrains the coefficients of non-causal relationships to zero. Let $\log L_0$ and $\log L_1$ denote the maximum log-likelihoods under models $\mathcal{M}_0$ and $\mathcal{M}_1$, respectively. The likelihood ratio test statistic is defined as: $\Lambda = -2(\log L_0 - \log L_1)$. The connection to the AIC difference is $\Delta\text{AIC} = \text{AIC}_0 - \text{AIC}_1 = \Lambda - 2\Delta k$ or equivalently $\Lambda = \Delta\text{AIC} + 2\Delta k$. Assuming $\mathcal{M}_0$ is the correct model for describing customer basket-shopping behavior, and under regularity conditions, the LRT statistic Λ approximately follows a chi-square distribution with degrees of freedom equal to the difference in the number of parameters: $\Lambda \sim \chi_{\Delta k}^2$ where $\Delta k = k_1 - k_0$.

This allows us to compute a p -value for testing whether the restrictions imposed by the causal model are consistent with the data: $p = \Pr(\chi_{\Delta k}^2 > \Lambda)$. Based on the above analysis, we compute the LRT statistic between the causal product network and alternative network structures for both Hypotheses 1 and 2.

Table 8 LRT Statistic Comparison (Hypothesis 1)

Theme	Best Benchmark	AIC (Best)	AIC (Model e)	ΔAIC	Δk	LRT Statistic
Theme 1	b2	559,393	557,721	1,672	261	2,194***
Theme 2	b1	437,961	436,085	1,876	460	2,796***
Theme 3	b2	308,765	304,645	4,120	217	4,554***
Theme 4	b2	646,568	638,588	7,980	188	8,356***
Theme 5	d	857,323	822,830	34,493	-118	34,729***

Note: (1) * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$; (2) Models a1 and a2 represent the Lasso-regularized complete network and the Ridge-regularized complete network, respectively; models b1 and b2 represent the SVD-decomposed correlation network and the EVD-decomposed correlation network; model c represents the collaborative filtering-based network; model d represents the association network; and model e represents the causal product network; (3) The best benchmark model refers to the model with the lowest AIC score for each theme; (4) The LRT statistic Λ approximately follows a chi-square distribution with degrees of freedom equal to the difference in the number of parameters.

Table 8 presents the likelihood ratio test results comparing the causal product network (model e) with the best benchmark model for each theme. Across all five themes, the causal product network consistently outperforms the best benchmark model in terms of both model fit and statistical significance. In Theme 1 (Pasta), the causal model improves upon the best benchmark (model b2) by reducing the AIC by 1,672, and although it uses 261 fewer parameters, the resulting LRT statistic is 2,194. It indicates strong evidence of improved fit. In Theme 2 (QSR), the causal model achieves an AIC improvement of 1,876 over model b1 and uses 460 fewer parameters, resulting in an LRT statistic of 2,796. The difference is even more pronounced in Theme 3 (Bakery), where the causal model improves AIC by 4,120 with 217 fewer parameters, yielding an LRT statistic of 4,554. In Theme 4 (Prepared Food), the causal model lowers the AIC by 7,980

compared to model b2 and uses 188 fewer parameters, resulting in an LRT statistic of 8,356. Finally, in Theme 5 (Unprepared Food), the causal model achieves the largest improvement, reducing the AIC by 34,493 relative to model d, despite using 118 more parameters; the resulting LRT statistic of 34,729 still indicates overwhelming support for the causal model. In all cases, the improvements are statistically significant at the 1% level, confirming the superior performance of the causal product network.

Table 9 LRT Statistic Comparison (Hypotheses 2)

Theme	Best Benchmark	AIC (Best)	AIC (Model e)	Δ AIC	Δk	LRT Statistic
Theme 1	h	561,288	557,721	3,567	92	3,751***
Theme 2	h	440,309	436,085	4,224	98	4,420***
Theme 3	h	309,806	304,645	5,161	81	5,323***
Theme 4	h	641,099	638,588	2,511	62	2,635***
Theme 5	h	881,728	822,830	58,898	92	59,082***

Note: (1) * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$; (2) Model e represents the causal product network; models f, g, and h represent the category-level causal effects model, the product-to-category causal effects model, and the category-to-product causal effects model, respectively; (3) The best benchmark model refers to the model with the lowest AIC score for each theme; (4) The LRT statistic Λ approximately follows a chi-square distribution with degrees of freedom equal to the difference in the number of parameters.

As shown in Table 9, the causal product network outperform other best alternative specification in terms of both model fit (AIC) and statistical validation (LRT) across all five themes. In Theme 1 (Pasta), it lowers the AIC by 3,567 relative to model h, with 92 additional parameters, resulting in an LRT statistic of 3,751. Similarly, for Theme 2 (QSR), the causal model achieves an AIC reduction of 4,224 and includes 98 more parameters, yielding an LRT value of 4,420. The improvement is more substantial in Theme 3 (Bakery), with a 5,161 drop in AIC and 81 extra parameters, leading to an LRT statistic of 5,323. In Theme 4 (Prepared Food), the causal model surpasses model h by decreasing the AIC by 2,511 while adding 62 parameters, resulting in an LRT score of 2,635. Theme 5 (Unprepared Food) shows the most dramatic result: a reduction of 58,898 in AIC with 92 additional parameters, producing an LRT statistic of 59,082. In every case, the observed improvements are highly statistically significant at the 1% level, reinforcing the causal product network’s effectiveness over other causal specifications.